QCon[上海站]
全球软件开发大会2016

Progressive Web Apps

Make The Web Great Again (复兴序章)

黄玄/@huxpro

@QCon Shanghai Oct 2016

Follow @huxpro 685 Star 0

huangxuan.me/pwa-qcon2016/



huangxuan.me/pwa-qcon2016

(推荐您访问网址观看)

24 Aug 2016 14:02



先考虑你有没有兴趣来qcon讲，讲师福利是可以免费听所有的内容，还报销机票和安排住宿。


"It's a trap!!"



```
export default () => (
  <Huxpro zh='黄玄' en='Hux'>
    <Wepiao from="2015" team="Frontend Infrastructure" />
    <Taobao from="2014" team="AliTrip - Mobile Web" />
  </Huxpro>
)
```

huangxuan.me / Weibo / GitHub / 知乎

Progressive Web App?

Let's first talk about

The State Of Web

World Wide Web

The WorldWideWeb (W3) is a wide-area hypermedia information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an executive summary of the project, Mailing lists , Policy , November's W3 news , Frequently Asked Questions .

What's out there?
: Pointers to the world's online information, subjects , W3 servers, etc.

Help
: on the browser you are using

Software Products
: A list of W3 project components and their current state. (e.g. Line Mode ,X11 Viola , NeXTStep , Servers , Tools , Mail robot , Library)

Technical
: Details of protocols, formats, program internals etc

Bibliography
: Paper documentation on W3 and references.

People
: A list of some people involved in the project.

History
: A summary of the history of the project.

How can I help ?
: If you would like to support the web..

Getting code
: Getting the code by anonymous FTP , etc.

"Web has progressed from a large universe of documents to a universal application platform."

The Progress of Web Apps - Microsoft Edge Team


WE WON ON
DESKTOP!

"The Strength of (Desktop) Web"

URL

Easy Discovery
Linkable

Open

Cross Platform
Broad Reach

Capable

Performant
Intergration

On-Demand

Low Friction
Seamless Release

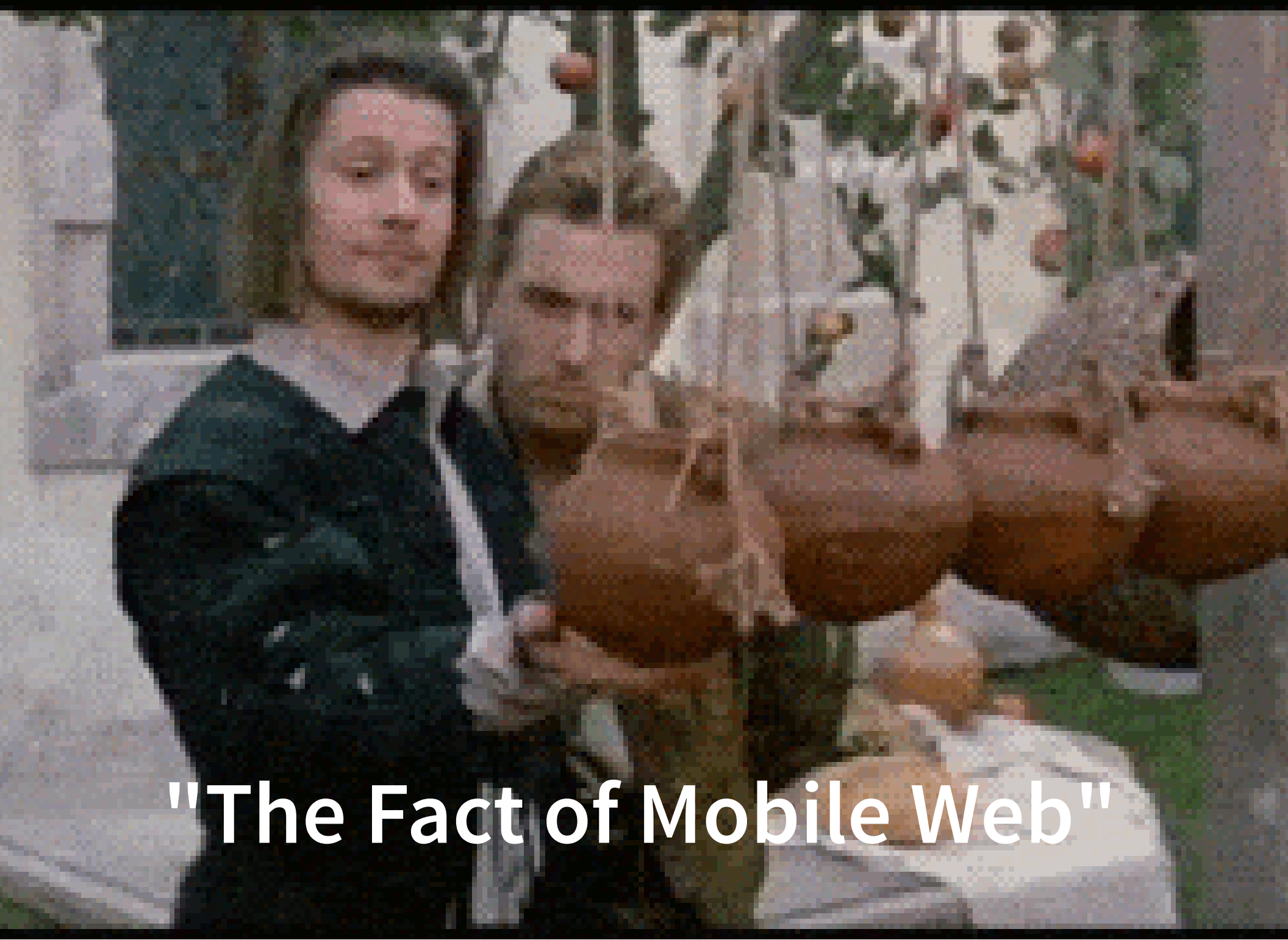

"...JUST IN TIME
FOR MOBILE TO
EAT THE WORLD"
CODE
AMERICA
Google
Accessibility
whats-next-for-the-web-platform-alex-russell

"DON'T WORRIED"



"WE HAVE HTML5!"


"New open standards created in the mobile era, such as HTML5, will win on mobile devices." (2010)
http://www.apple.com/hotnews/thoughts-on-flash/


"The Fact of Mobile Web"

"Web On Mobile"

URL

Easy Discovery
Linkable

Open

Cross Platform
Broad Reach

Capable

~~Performant~~
~~Intergration~~

On-Demand

Low Friction
Seamless Release

Low capability

Web

↓

↓

↓

Lacking...

↓

↓

↓

Native

High capability

Performance

UI Performance

DOM & Retained Mode/GPU Rasterization

Network

Latency/Cache/Offline/Network Resilience

Intergration

Device Access

Media/Geolocation/Gesture/Accelerometer/etc.

OS Intergration

HCI/HomeScreen/Push Notification/FileSystem/

Background Threads/etc.


"I think the biggest mistake that we made, as a company, is betting too much on HTML5 as opposed to native ." (2012)
https://techcrunch.com/2012/09/11/mark-zuckerberg-...-html5/

PPK: Web vs. native: let's concede defeat (2015)

WIRED: The Web Is Dead. Long Live the Internet (2010)

WSJ: The Web Is Dying; Apps Are Killing It (2013)

Why mobile web is slow (2013)

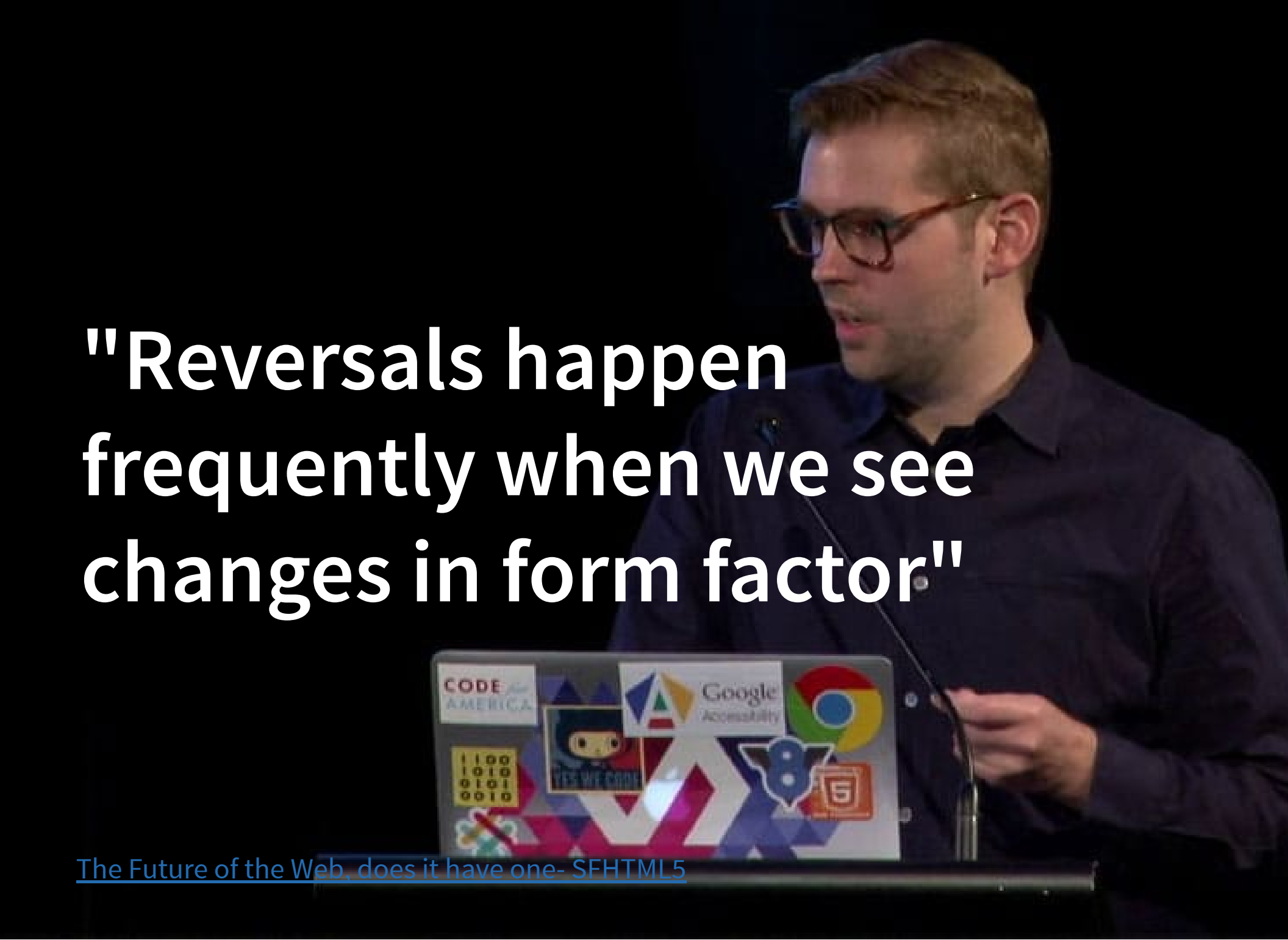
TC: As Mobile Roars Ahead, It's Time To Finally Admit The Web Is Dying (2014)

TheVerge: The mobile web sucks (2015)

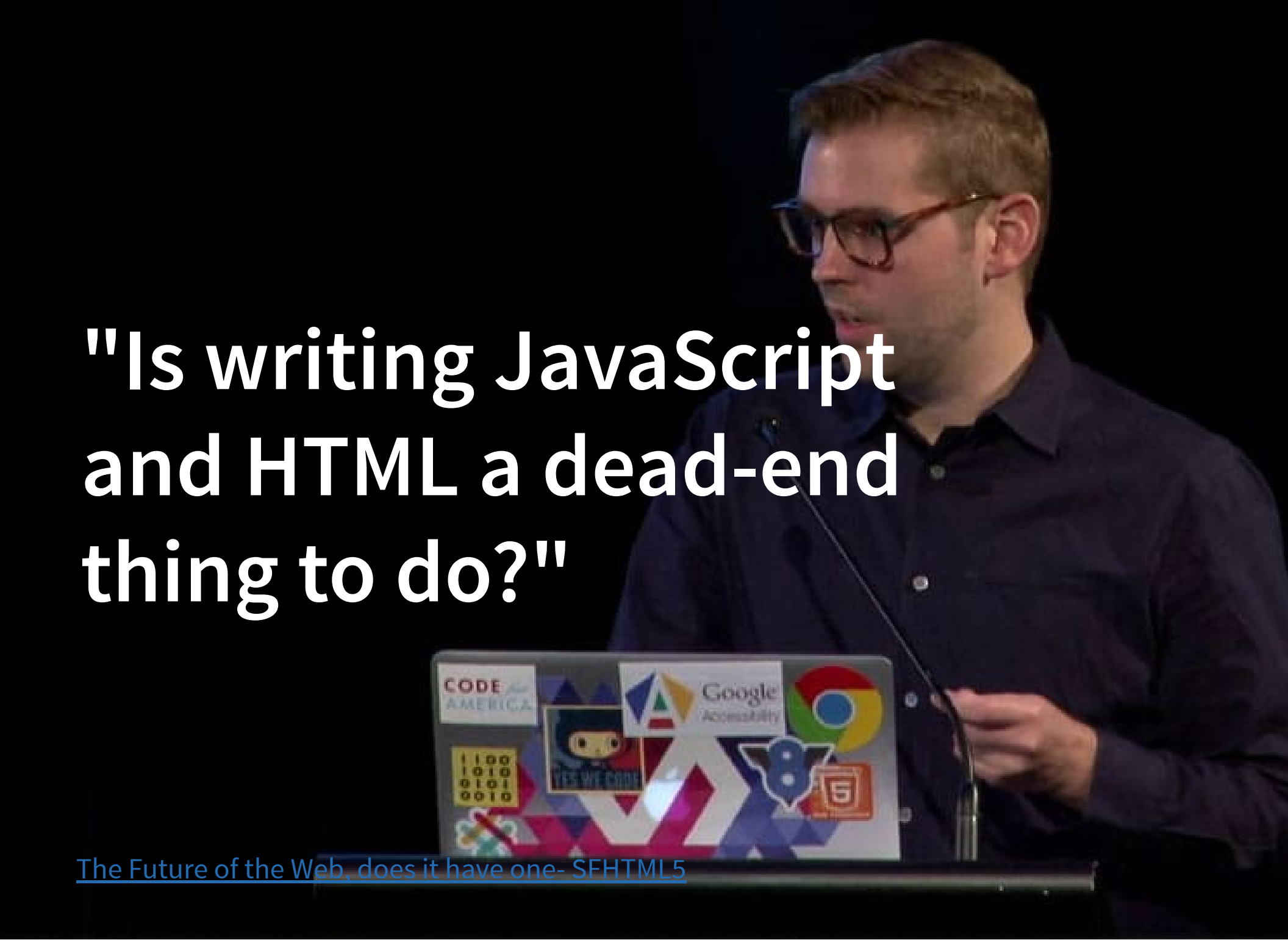
Living With Web Apps: the 10 deadly sins of a mobile web app (2015)

The Fucking Open Web (2016)

Mobile JavaScript Apps: The Problem with the Mobile Web (2016)



The Future of the Web, does it have one- SFHTML5



The Future of the Web, does it have one- SFHTML5

Community Stick On Web and Push...

Web Technologies


"we feel the web as a platform is at a disadvantage, and we're working to polyfill those gaps with PhoneGap." (2008)
http://phonegap.com/blog/2012/05/09/phonegap-beliefs-goals-and-philosophy/

Low capability

Web

↓

↓

↓

Hybrid

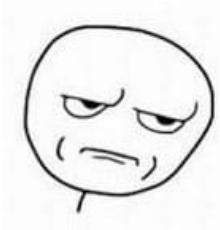
↓

↓

↓

Native

High capability



Performance

UI Performance ***(We can make it!)***

DOM & Retained Mode/GPU Rasterization

✓ Network

Latency/Cache/Offline/Network Resilience

✓ **Intergration**

✓ Device Access

Media/Geolocation/Gesture/Accelerometer/etc.

✓ OS Intergration

HCI/HomeScreen/Push Notification/FileSystem/

Background Threads/etc.

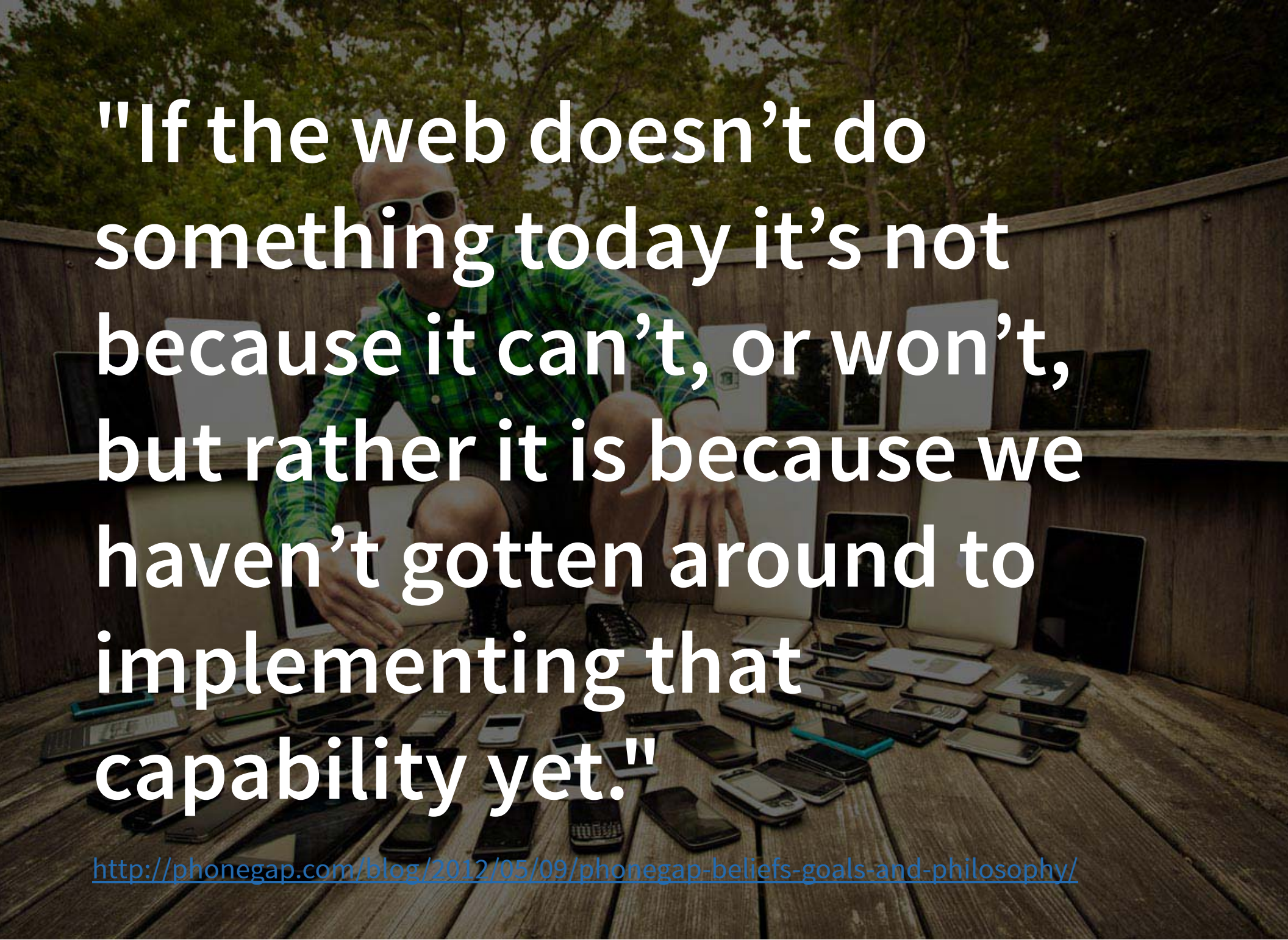
Tons of "Hybridzation"



Hybridzation Toke Our Soul, Put It Back Into Garden Wall.

Nobody, maybe me


"We believe the web has been the most convincing solution to reaching many devices of differing capabilities."
http://phonegap.com/blog/2012/05/09/phonegap-beliefs-goals-and-philosophy/


"If the web doesn’t do something today it’s not because it can’t, or won’t, but rather it is because we haven’t gotten around to implementing that capability yet."
http://phonegap.com/blog/2012/05/09/phonegap-beliefs-goals-and-philosophy/

We already walked too far,
down to we had forgotten why embarked.




"PhoneGap is a Polyfill, and the ultimate purpose of PhoneGap is to cease to exist." (2012)
http://phonegap.com/blog/2012/05/09/phonegap-beliefs-goals-and-philosophy/

Time To Esc Walled Garden, And Back To Open Free Land.

Nobody, maybe me


"Progressive Web Apps: Escaping Tabs Without Losing Our Soul" (2015)
CODE
AMERICA
Google
Accessibility
progressive-apps-escaping-tabs-without-losing-our-soul/

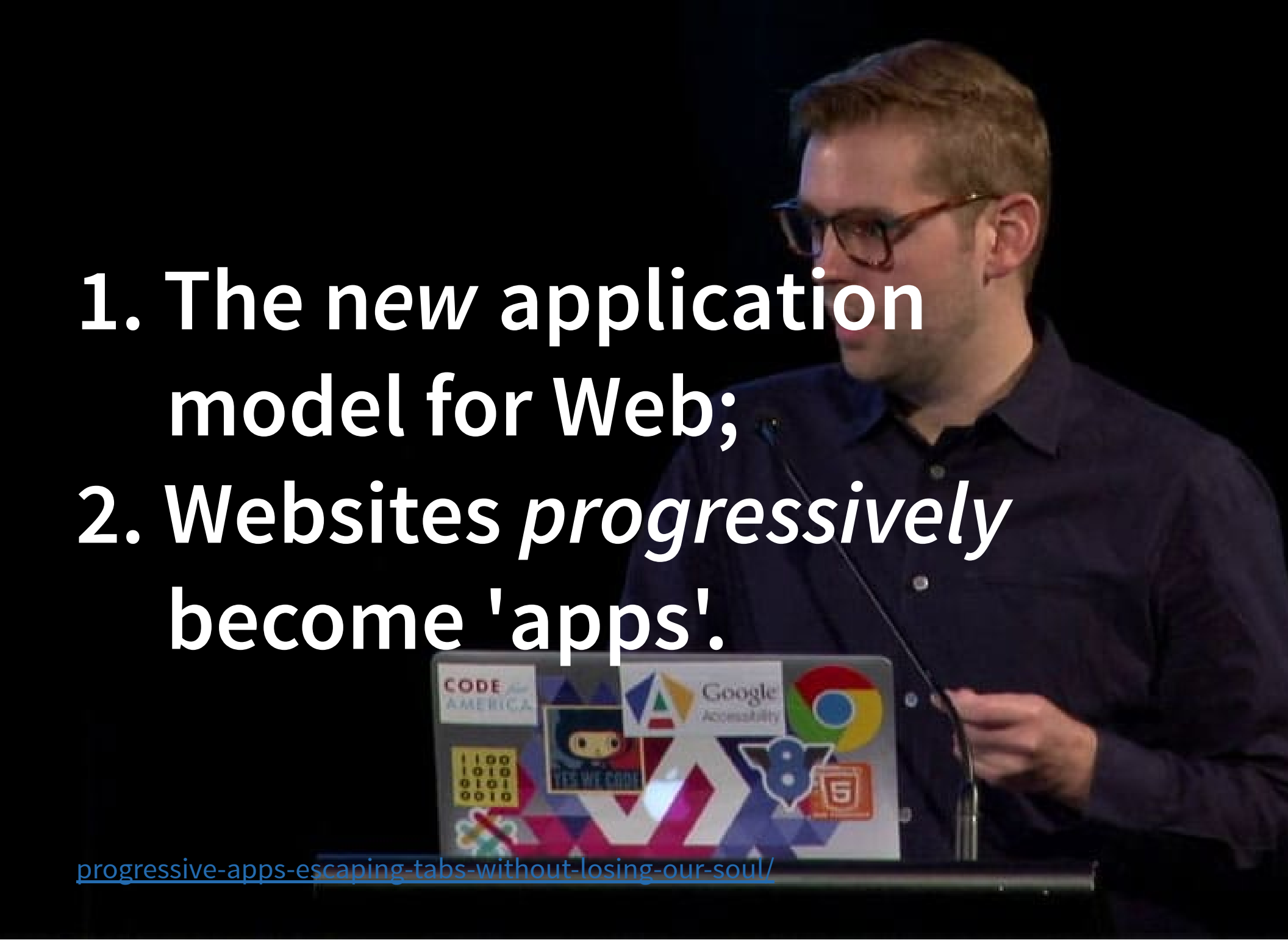
Introduction to PWA: From What to How

PWA 101

progressive
adjective

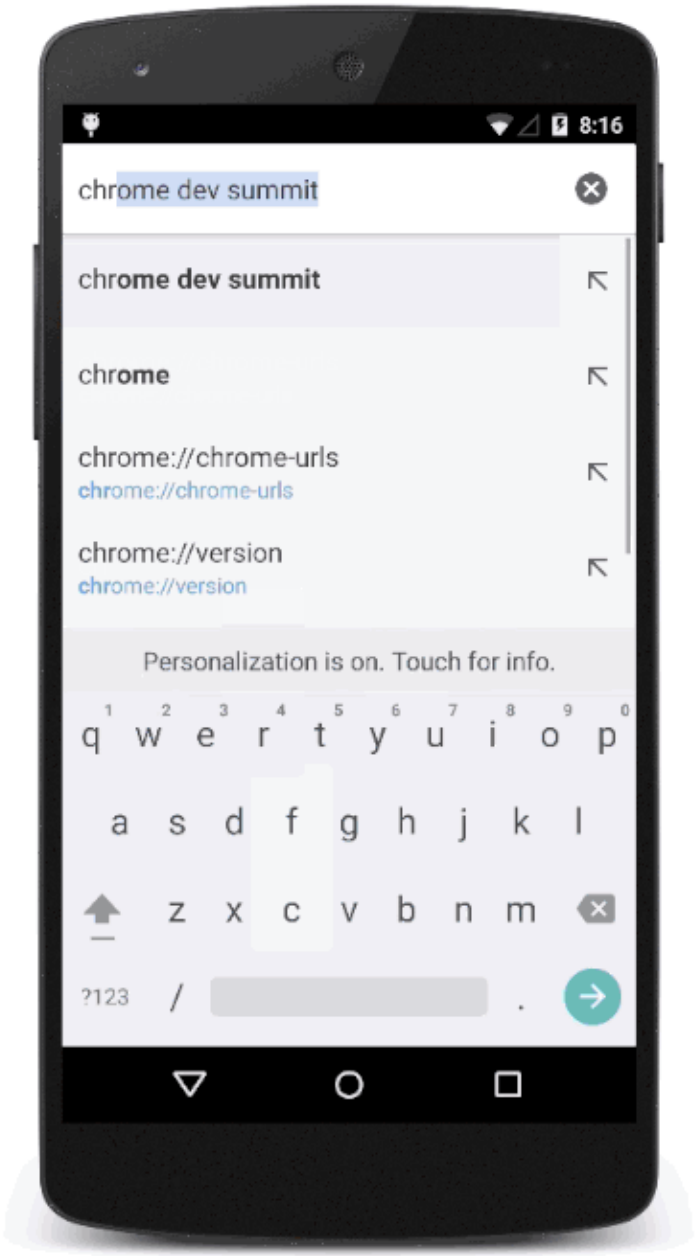
1 *the progressive deterioration of the social conditions of farm labourers*: **CONTINUING,** continuous, increasing, growing, developing, ongoing, intensifying, accelerating, escalating; gradual, step by step, cumulative.

2 *a teacher with progressive views on primary education*: **MODERN,** liberal, advanced, forward-looking, forward-thinking, go-ahead, enlightened, enterprising, innovative, up-and-coming, new, dynamic, avant-garde, modernistic, disruptive; radical, left-wing, reforming, reformist, revolutionary, revisionist, progressivist. ANTONYMS conservative, reactionary.


1. The new application model for Web;
2. Websites progressively become 'apps'.
CODE for AMERICA
Google Accessibility
progressive-apps-escaping-tabs-without-losing-our-soul/

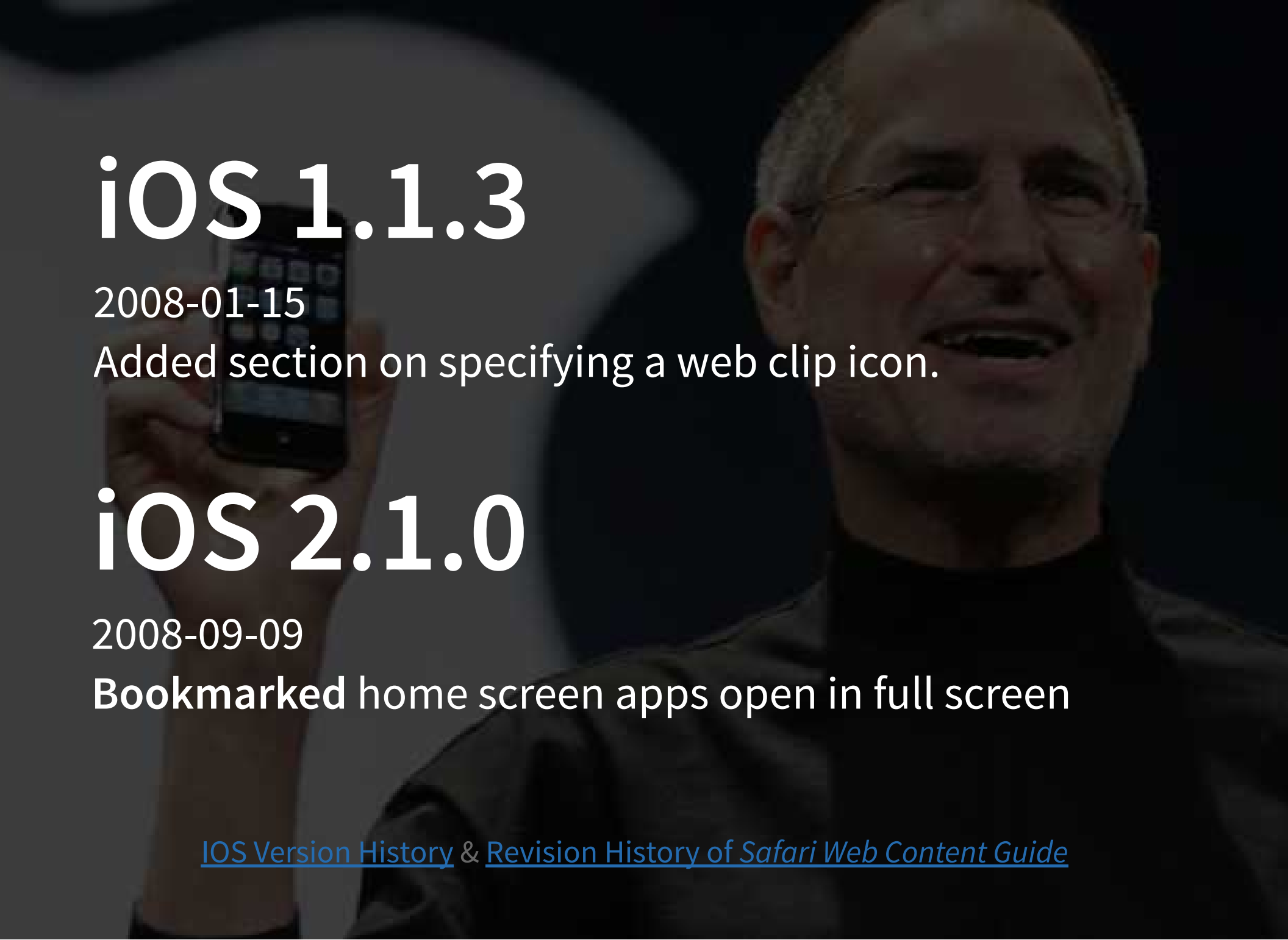
No.1, The Beginning of First-Class Citizen

Add To HomeScreen



Chrome Dev Summit 2014 [github]





iOS 1.1.3

2008-01-15

Added section on specifying a web clip icon.

iOS 2.1.0

2008-09-09

Bookmarked home screen apps open in full screen

IOS Version History & Revision History of *Safari Web Content Guide*

The "Good Old Time"

```
<!-- Add to homescreen for Chrome on Android -->
<meta name="mobile-web-app-capable" content="yes">
<mate name="theme-color" content="#000000">

<!-- Add to homescreen for Safari on iOS -->
<meta name="apple-mobile-web-app-capable" content="yes">
<meta name="apple-mobile-web-app-status-bar-style" content="bla
<meta name="apple-mobile-web-app-title" content="Lighten">

<!-- Tile icon for Win8 (144x144 + tile color) -->
<meta name="msapplication-TileImage" content="images/touch/ms-t
<meta name="msapplication-TileColor" content="#3372DF">

<!-- iOS icons -->
<link rel="apple-touch-icon-precomposed" sizes="144x144" href="
<link rel="apple-touch-icon-precomposed" sizes="114x114" href="
<link rel="apple-touch-icon-precomposed" sizes="72x72" href="im
<link rel="apple-touch-icon-precomposed" href="images/touch/app

<!-- Generic Icon -->
<link rel="shortcut icon" href="images/touch/touch-icon-57x57.p

<!-- Chrome Add to Homescreen -->
<link rel="shortcut icon" sizes="196x196" href="images/touch/to
```

Web App Manifest (SPEC)

```
{
  "short_name": "Manifest Sample",
  "name": "Web Application Manifest Sample",
  "icons": [{
      "src": "launcher-icon-2x.png",
      "sizes": "96x96",
      "type": "image/png"
   }],
  "background_color": "#fff",
  "theme_color": "#000",
  "start_url": "/index.html",
  "display": "standalone",
  "orientation": "landscape"
}
```

```
<link rel="manifest" href="/manifest.json">
```

https://developer.mozilla.org/en-US/docs/Web/Manifest

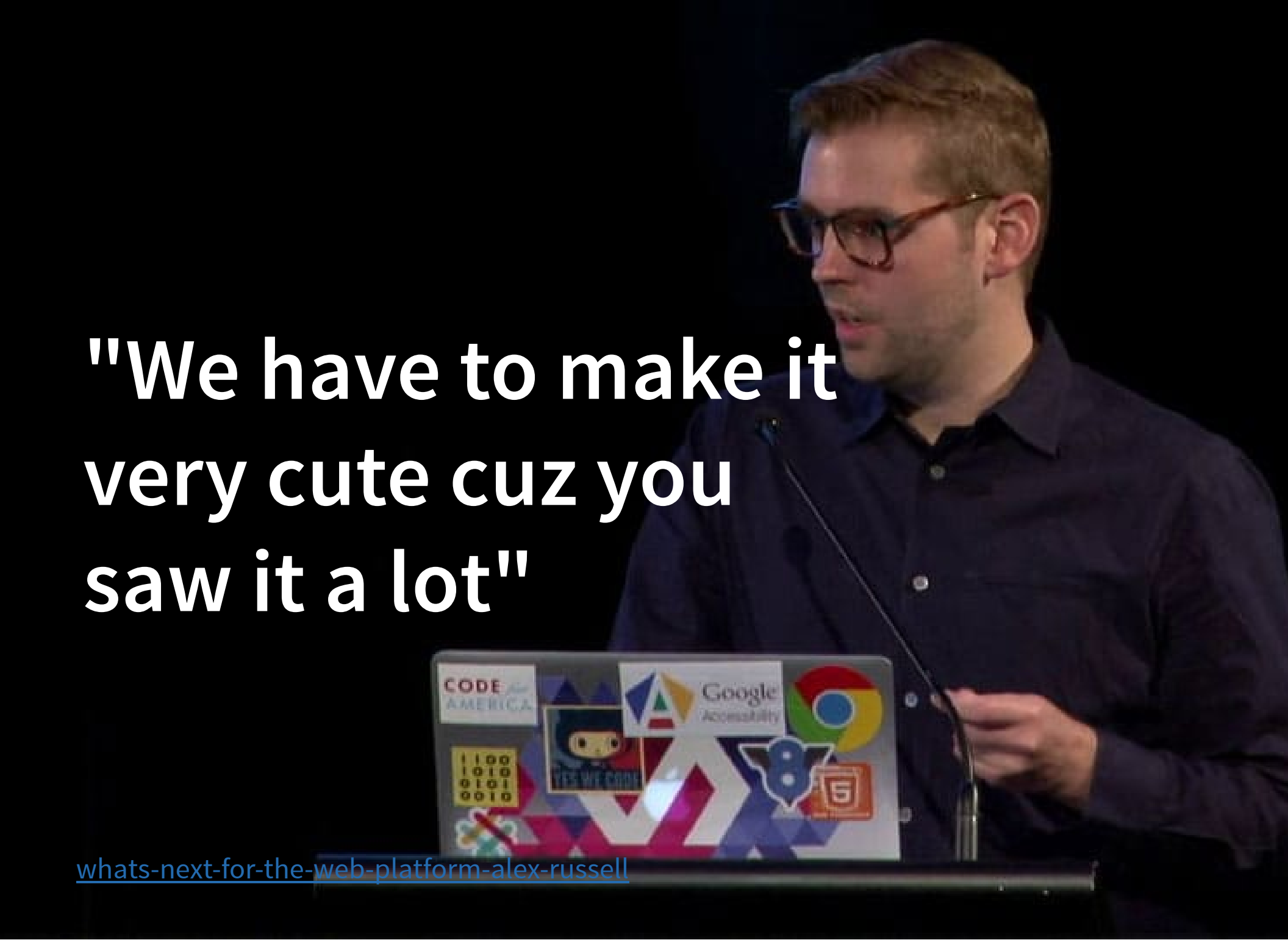
No.2 Network As Progressive Enhancement

Reliable Experience

(THIS IS A BROWSER, ACTUALLY)







whats-next-for-the-web-platform-alex-russell

Web is constructed with the assumption that you have put yourself into the "internet mode"

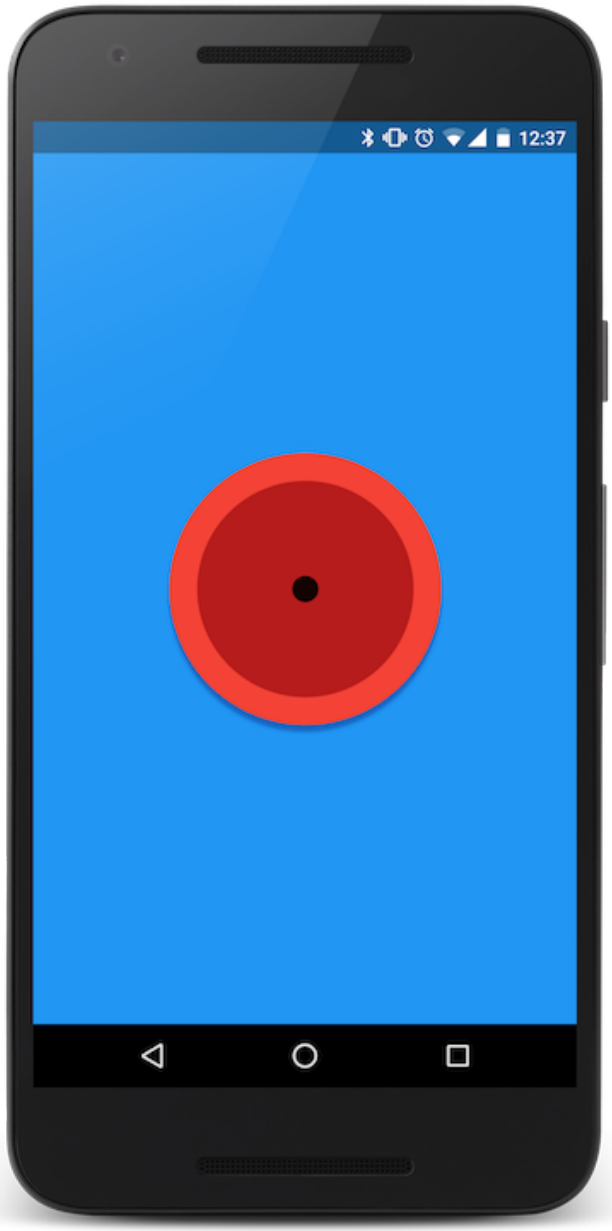
APP CACHE?

(Offline Web App Spec)

```
<html manifest="cache.appcache">
```


APPCACHE: DOUC
The "Onion"
http://alistapart.com/article/application-cache-is-a-douchebag
JSC
ARIZONA

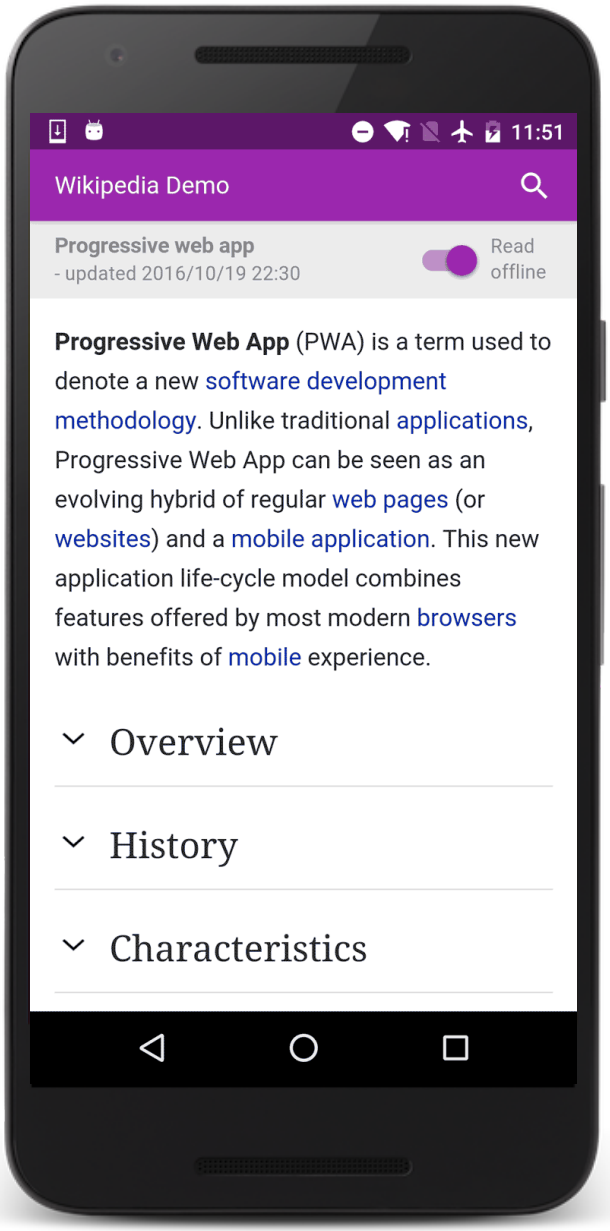

"等了这么久，现在有了
Service Worker 终于可以
do it properly."
The Kentucky Center
In private session ;)



SW Showcase

Reliable Experience

Live ShowCase: Airhorner.com



SW Showcase

Caching On-the-fly

[Live Showcase: Offline Wikipedia]

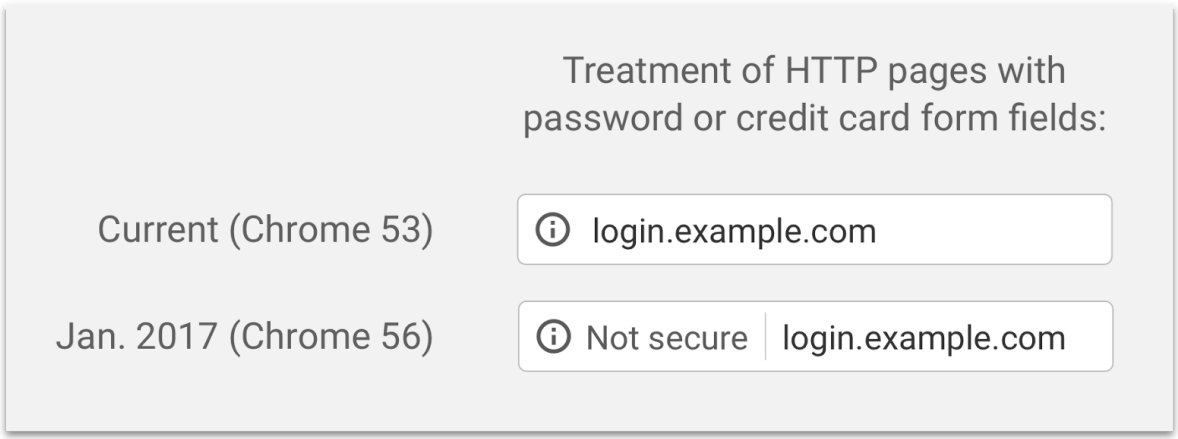

JS
THE SERVICEWORKER
@jaffathecake

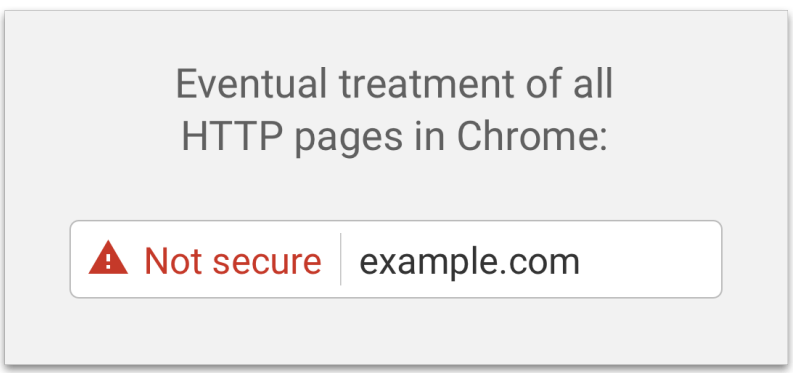
Service Worker (SPEC)

- Web Workers
- SharedWorker
- Chrome Background Pages
- Chrome Event Pages
- !App Cache

https://w3c.github.io/ServiceWorker/#motivations

HTTPS as Pre-requisite





[Moving Towards a More Secure Web - Chromium Blog]

Let's Build An AirHornor

Step1 - Register Service Worker

```
// register Service Worker in index.html
if('serviceWorker' in navigator) {
  navigator.serviceWorker
    .register('/sw.js')
    .then( registration => {
      console.log('Service Worker Registered');
    }).catch( error => {
      console.log('Registration failed with' + error);
    })
}
```

Step2 - On Install & Cache API(SPEC)

```
// sw.js
self.addEventListener('install', e => {
  e.waitUntil(
    caches.open('airhorner').then(cache => {
      return cache.addAll([
        '/',
        '/index.html',
        '/index.html?homescreen=1',
        '/?homescreen=1',
        '/styles/main.css',
        '/scripts/main.min.js',
        '/sounds/airhorn.mp3'
      ])
      .then(() => self.skipWaiting());
    })
  )
});
```

You manage what is 'installed' and how it's cached

Tips - ExtendableEvent (SPEC)

```
// WEBIDL
interface ExtendableEvent : Event {
  void waitUntil(Promise<any> f);
};
```

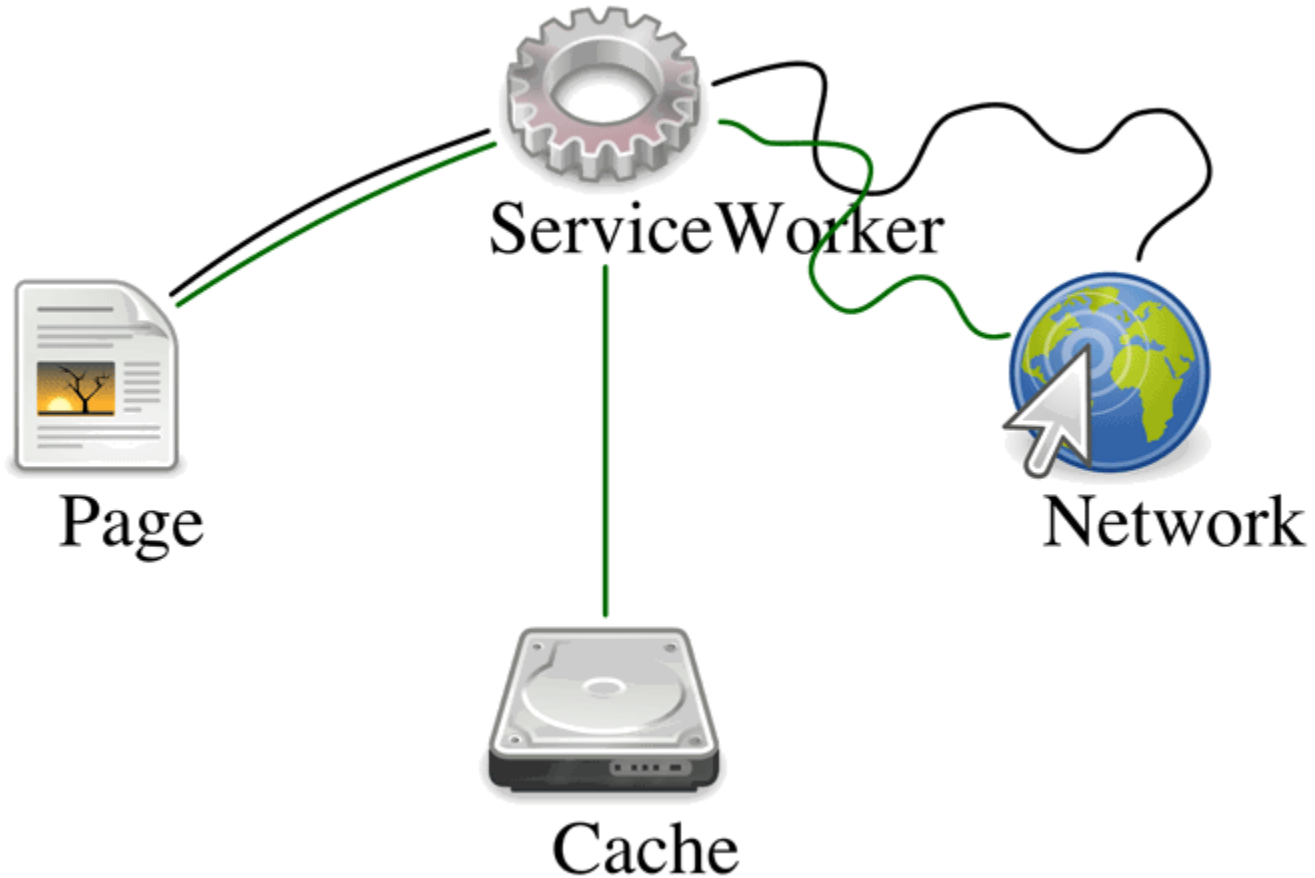
Tips - SkipWaiting (SEPC)

```
// WEBIDL
interface ServiceWorkerRegistration : EventTarget {
  readonly attribute ServiceWorker? installing;
  readonly attribute ServiceWorker? waiting;
  readonly attribute ServiceWorker? active;
}
```

Step3 - On Fetch

```
self.addEventListener('fetch', event => {
  event.respondWith(
    caches.match(event.request).then(response => {
      return response || fetch(event.request);
    })
  );
});
```

- You control request routing
- You control the caches
- You control the response



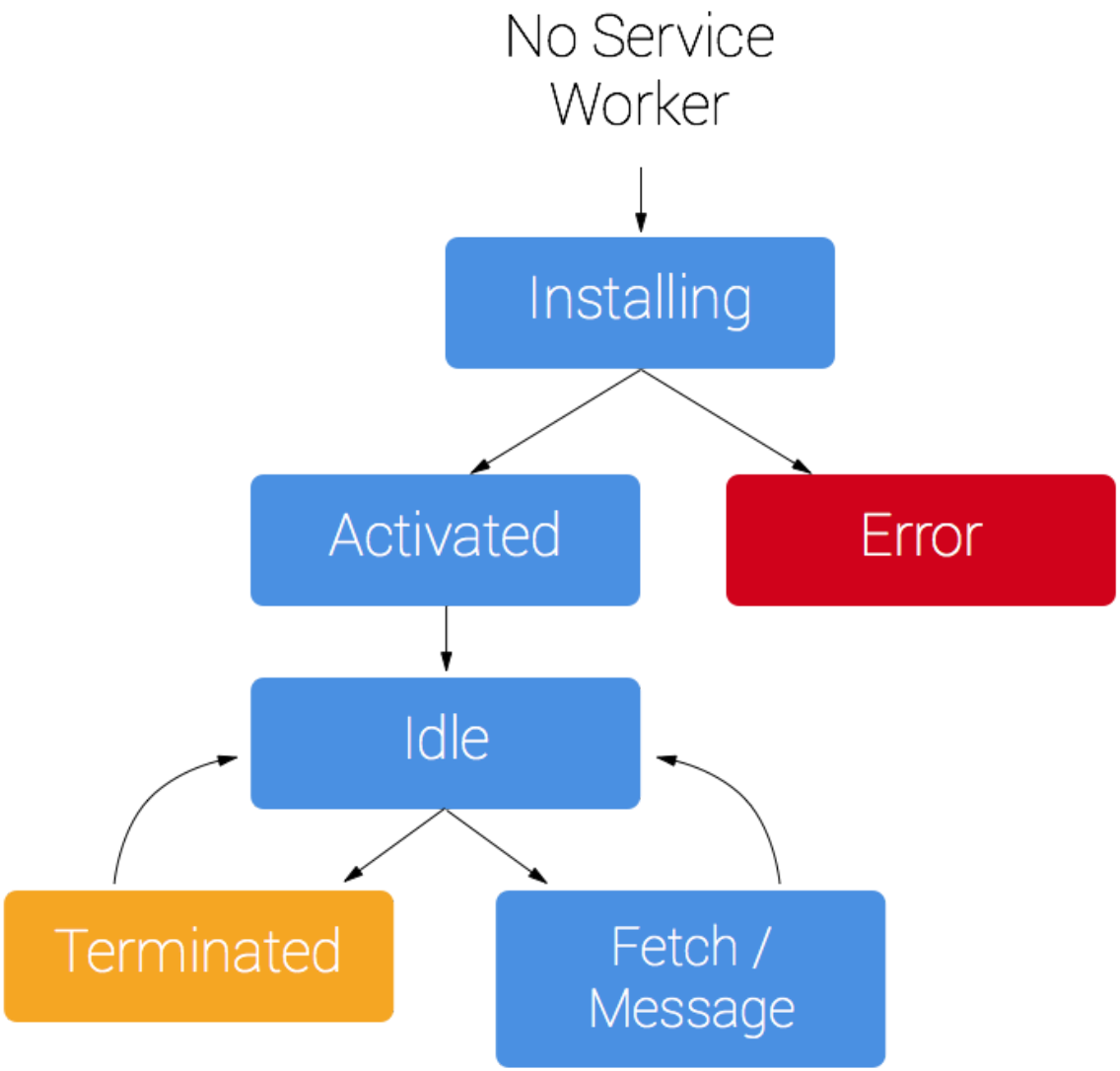
Let's It! - the Proxy Servers

Stale-While-Revalidate & Fallback

```
// sw.js
self.addEventListener('fetch', event => {
  event.respondWith(
    caches.open('mysite-dynamic').then( cache => {
      return cache.match(event.request).then( cachedResponse =>
        var fetchPromise = fetch(event.request)
          .then( networkResponse => {
            cache.put(event.request, networkResponse.clone());
            return networkResponse;
          }).catch( err => {
            return caches.match('offline.html')
          }
        return cachedResponse || fetchPromise;
      })
    })
  );
});
```

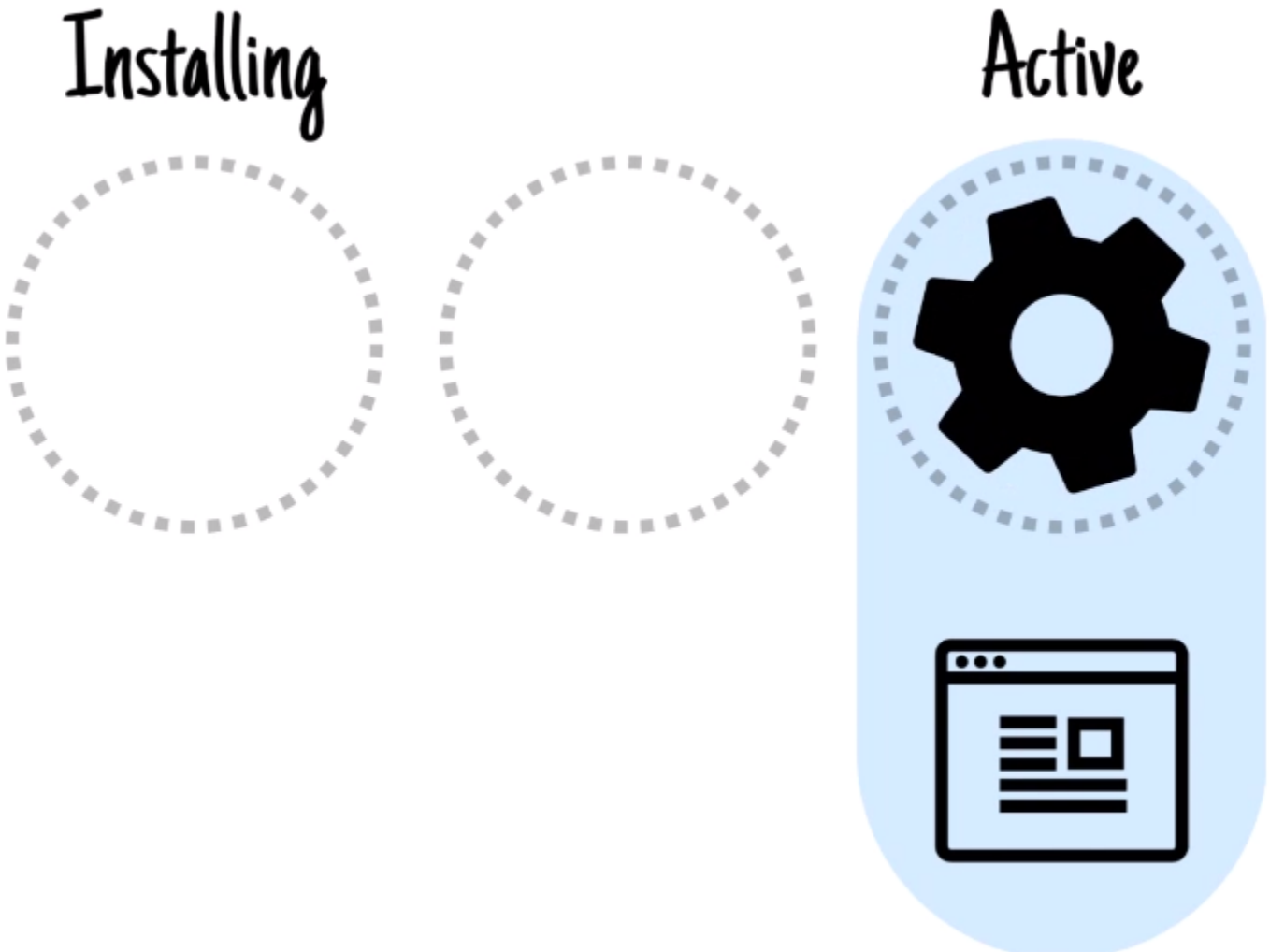
Hux Blog

Update SW?



LifeCycle

[HTML5Rocks] Introduction to Service Worker →



The First Service Worker

https://developers.google.com/web/fundamentals/instant-and-offline/service-worker/lifecycle



Updating the Service Worker

https://developers.google.com/web/fundamentals/instant-and-offline/service-worker/lifecycle

On Activate

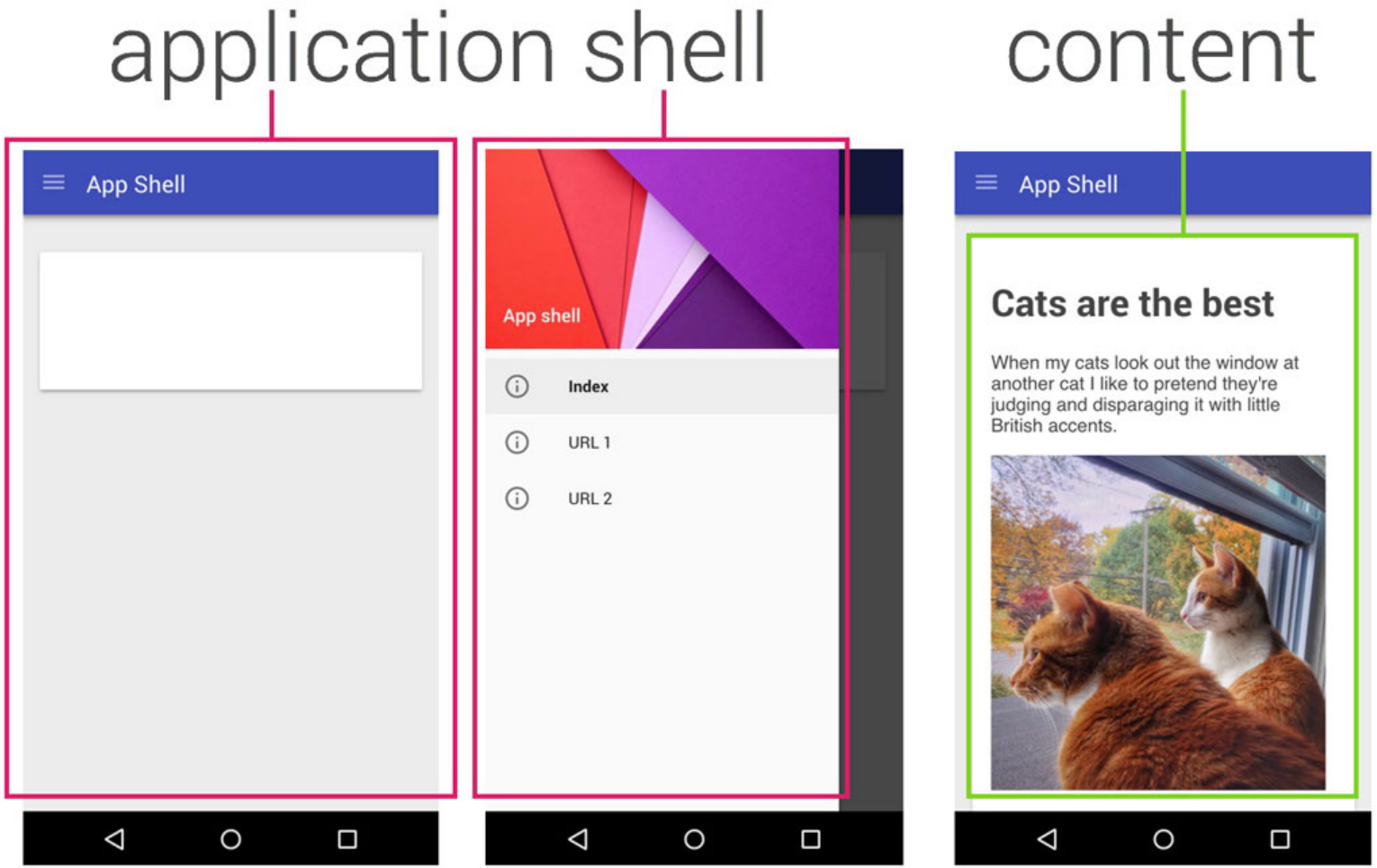
```
// sw.js
self.addEventListener('activate', function(event) {
  event.waitUntil(
    caches.keys().then(function(cacheNames) {
      return Promise.all(
        cacheNames.filter(function(cacheName) {
          // Return true if you want to remove this cache,
          // but remember that caches are shared across
          // the whole origin
        }).map(function(cacheName) {
          return caches.delete(cacheName);
        })
      );
    })
  );
});
```

Clean-up & Migration.

(SW can be built as progressive enhancement In.)

Architectural Revolution

Buzzword	Core Technology	Archtechutal Style
Ajax	**XHR**	**"Async First"**
RWD	**Media Query**	**"Mobile First"**
PWA	**Service Worker**	**"Offline First"**



Cached shell loads **instantly** on repeat visits.

Dynamic content then populates the view

Patterns & Libraries & More

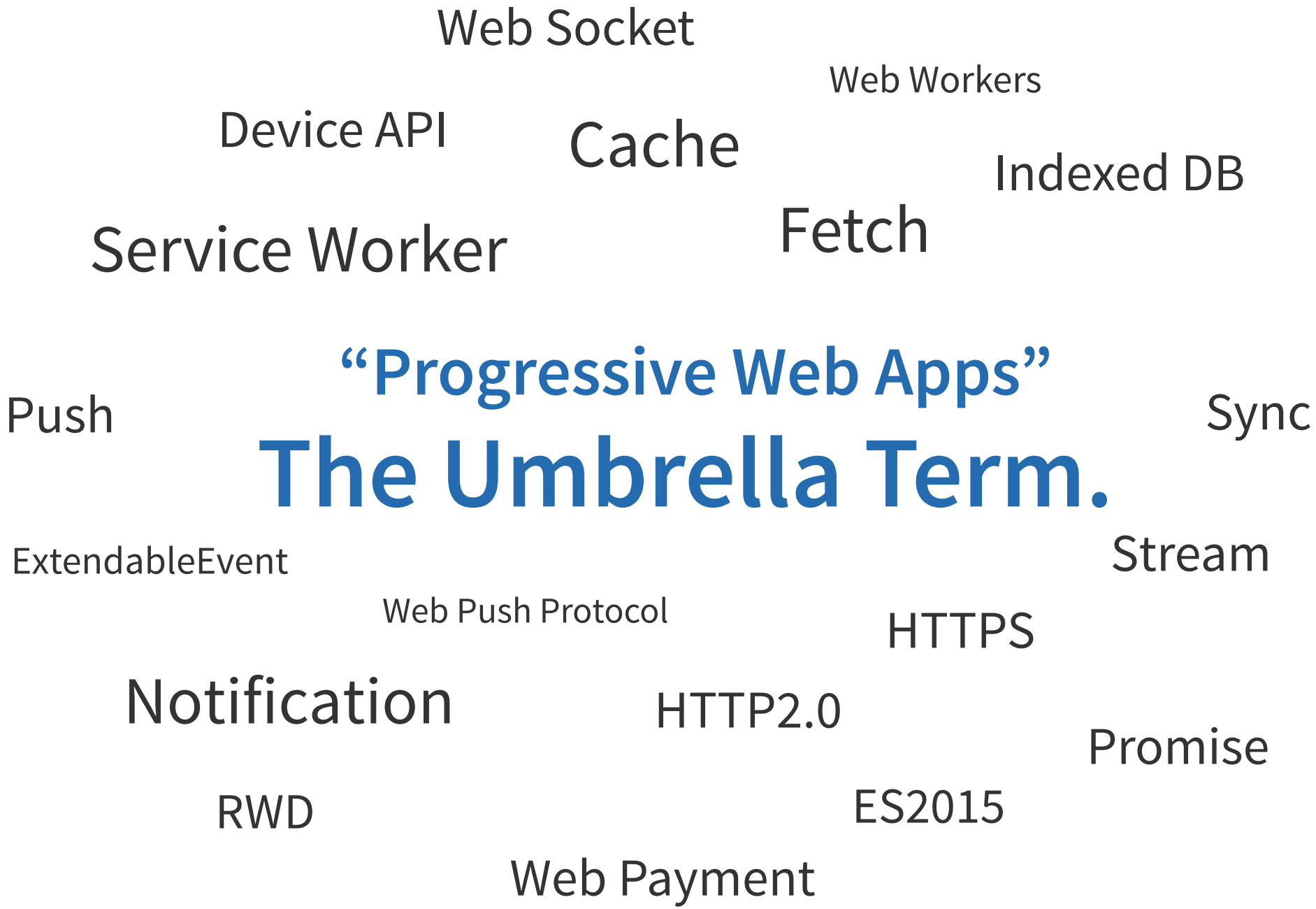
- App Shell Model
- Progressive Web App Summit 2016[Youtube]
- Web and Chrome at Google I/O 2016[Youtube]
- The offline cookbook
- Caching best practice & max-age
- GoogleChrome/sw-prefetch
- GoogleChrome/sw-toolbox

No.3 The Killer Features of Mobile OS

Re-engageable

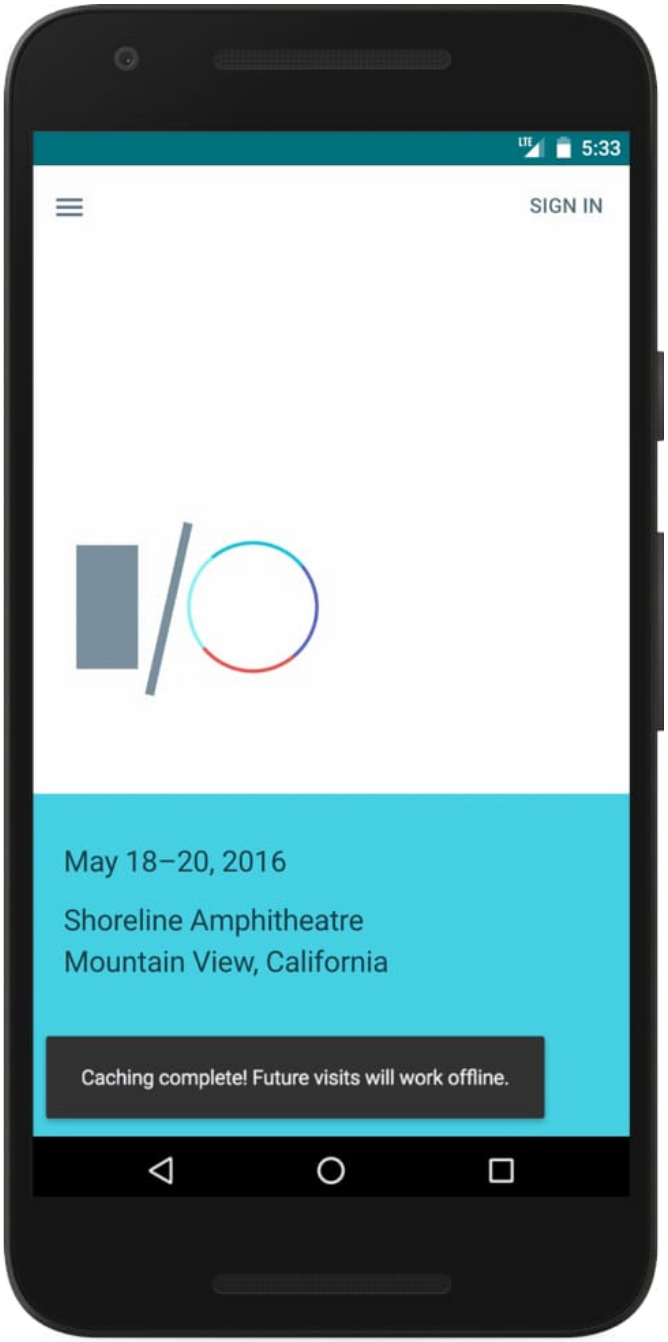
- Service Worker
- Push API & Notification API
- Background Sync
- Google Cloud Messaging (GCM/FCM)
- Web Push Protocol

push-notifications-on-the-open-web →


Web Socket
Web Workers
Device API
Cache
Indexed DB
Service Worker
Fetch
“Progressive Web Apps”
Push
Sync
The Umbrella Term.
ExtendableEvent
Stream
Web Push Protocol
HTTPS
Notification
HTTP2.0
Promise
RWD
ES2015
Web Payment

From UED' Pespective

PWA In Production

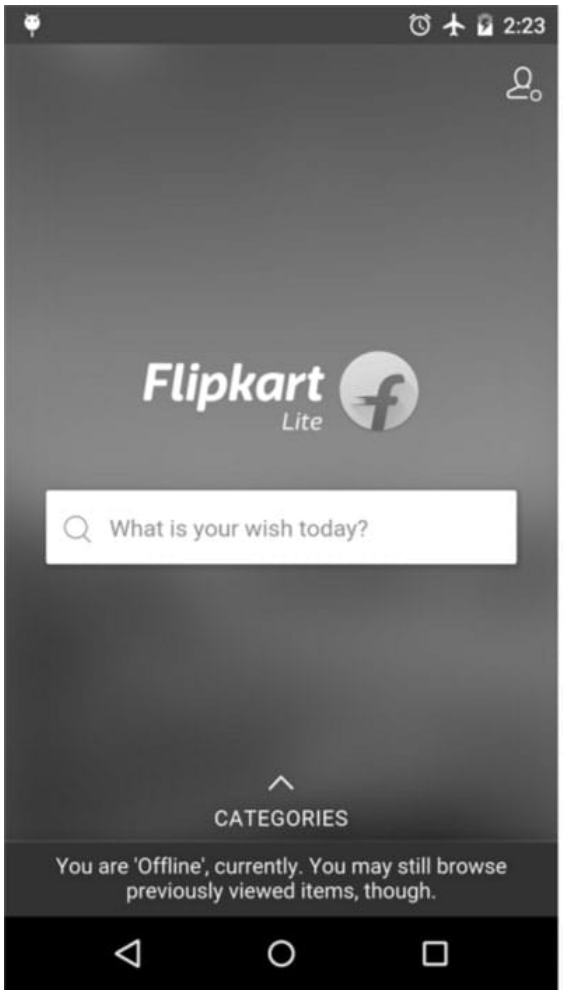

5:33
SIGN IN
May 18–20, 2016
Shoreline Amphitheatre
Mountain View, California
Caching complete! Future visits will work offline.



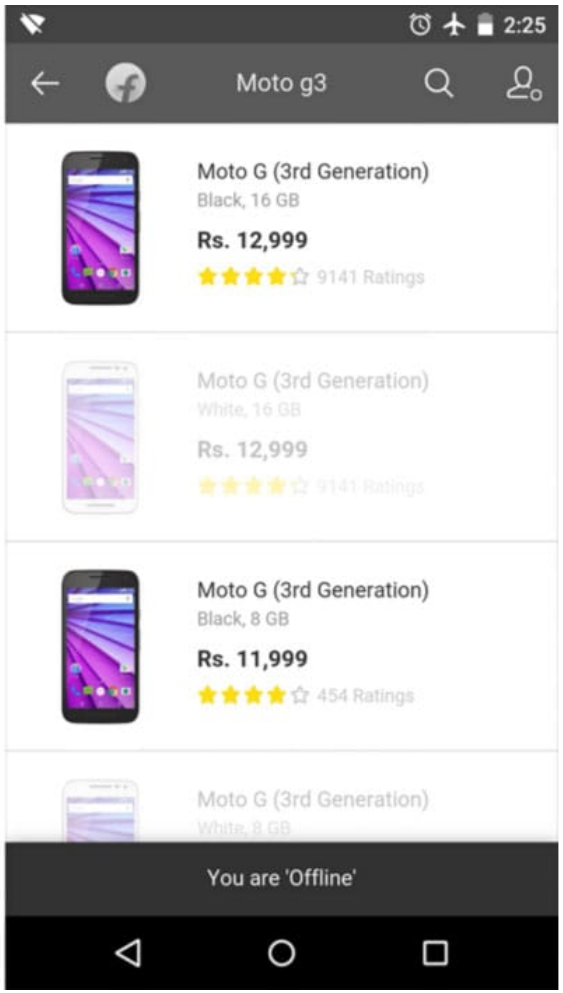
Visit Google in China

(without bypassing GFW)

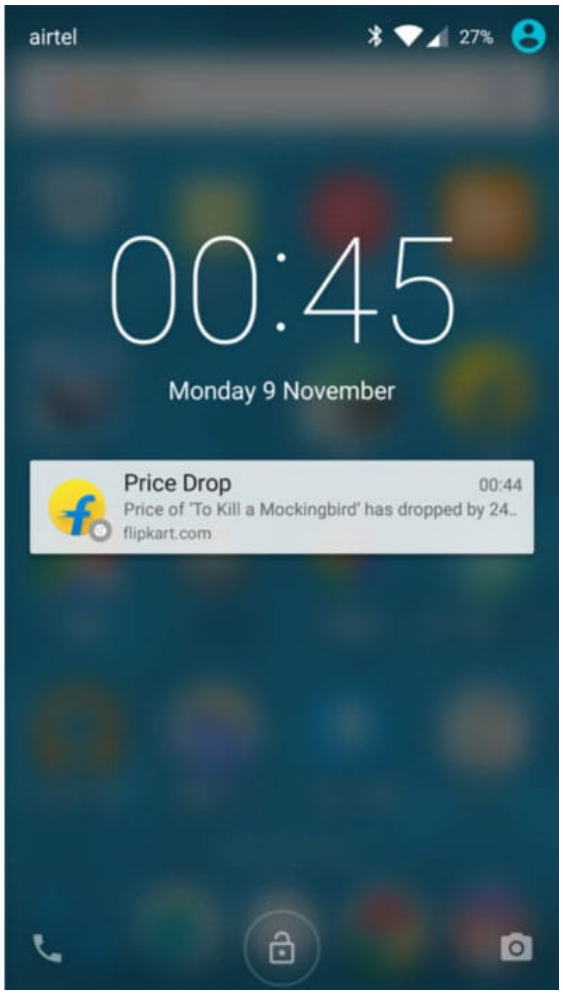
We Got Modern Web Capabilities



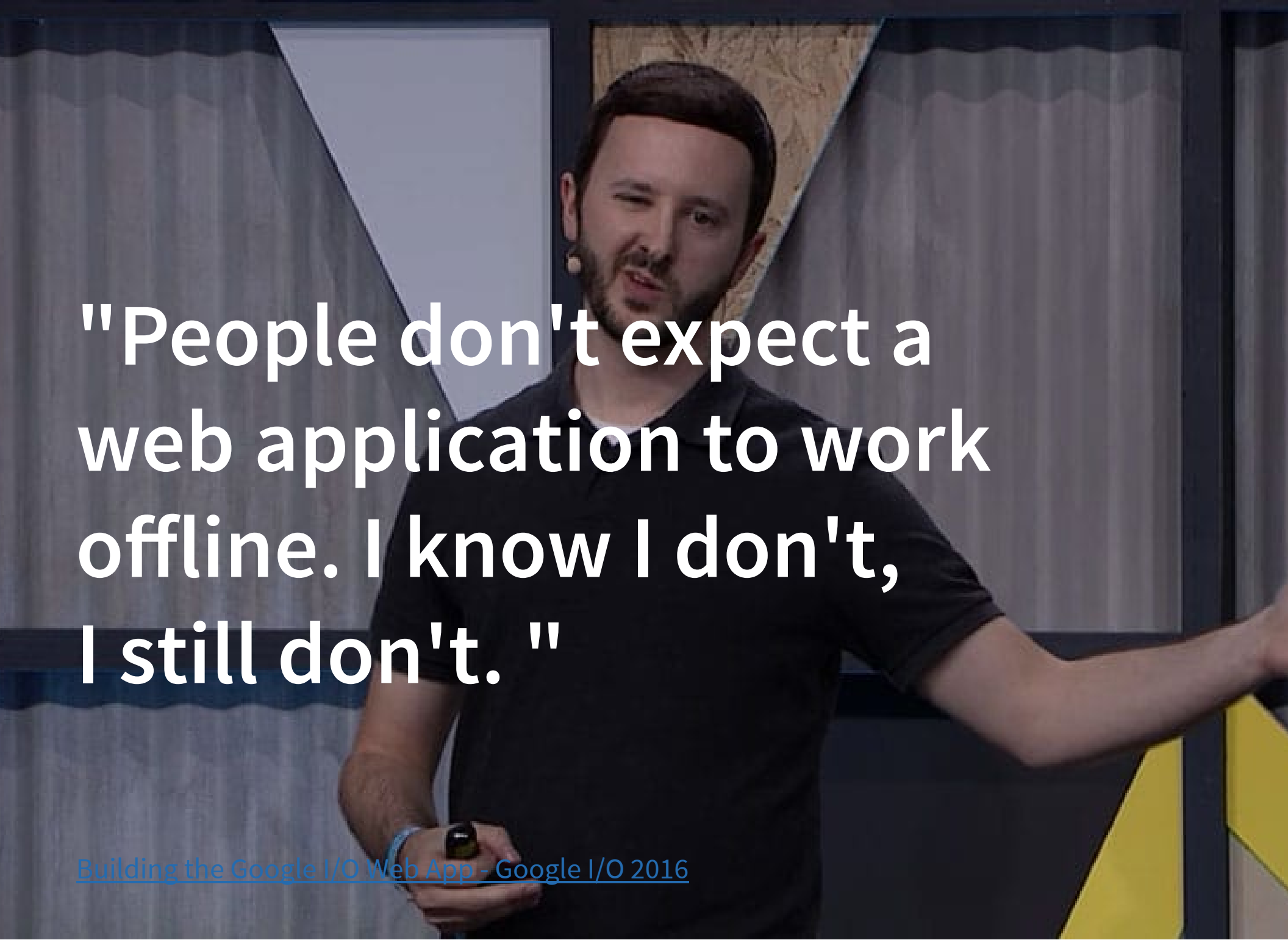
Work Offline



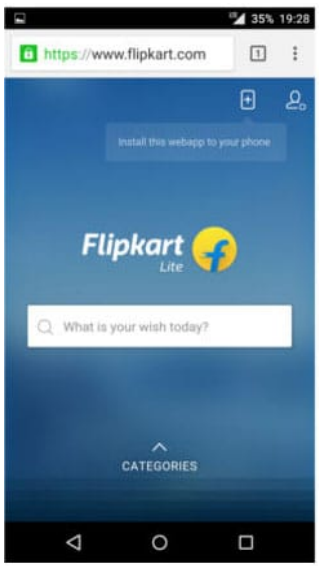
Caching

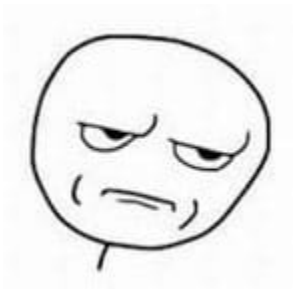


Push Notification

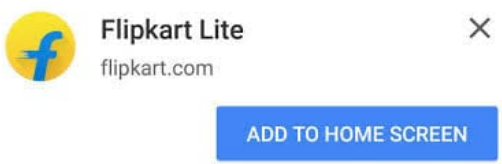

"People don't expect a
web application to work
offline. I know I don't,
I still don't. "
Building the Google I/O Web App - Google I/O 2016

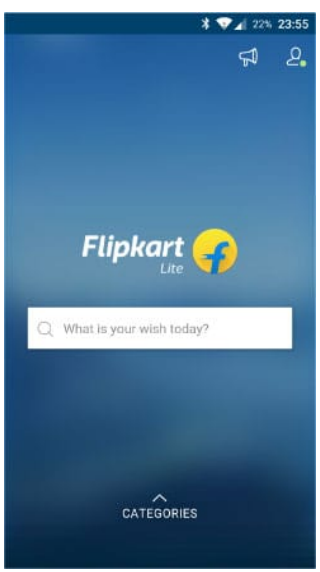
What we assume user would think





"Hey I saw the URL bar,
This is a **website**!"

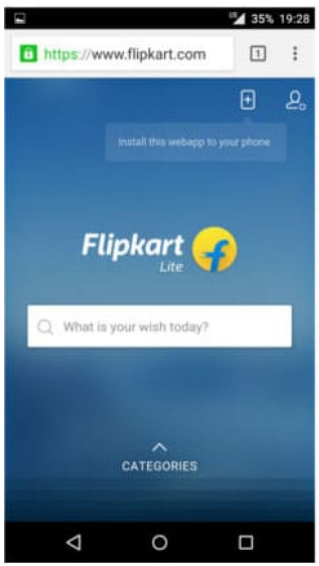


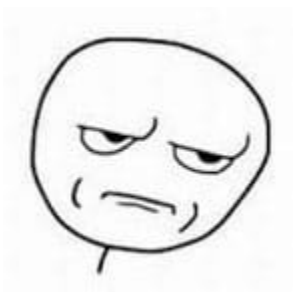




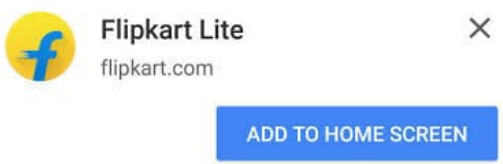
"This is an offline
reliable **app**!"

What user actually think

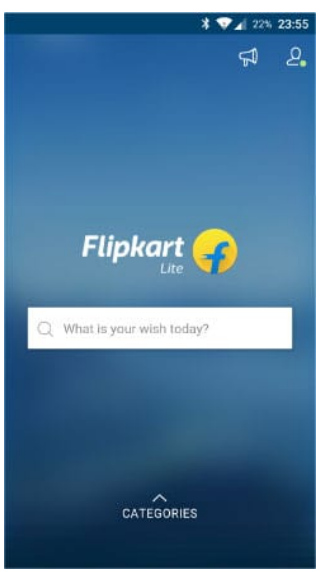


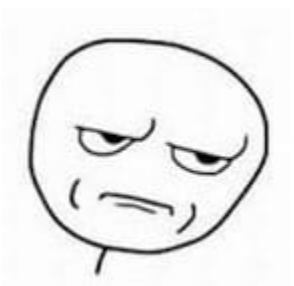


"Hey I saw the URL bar,
This is a **website**!"

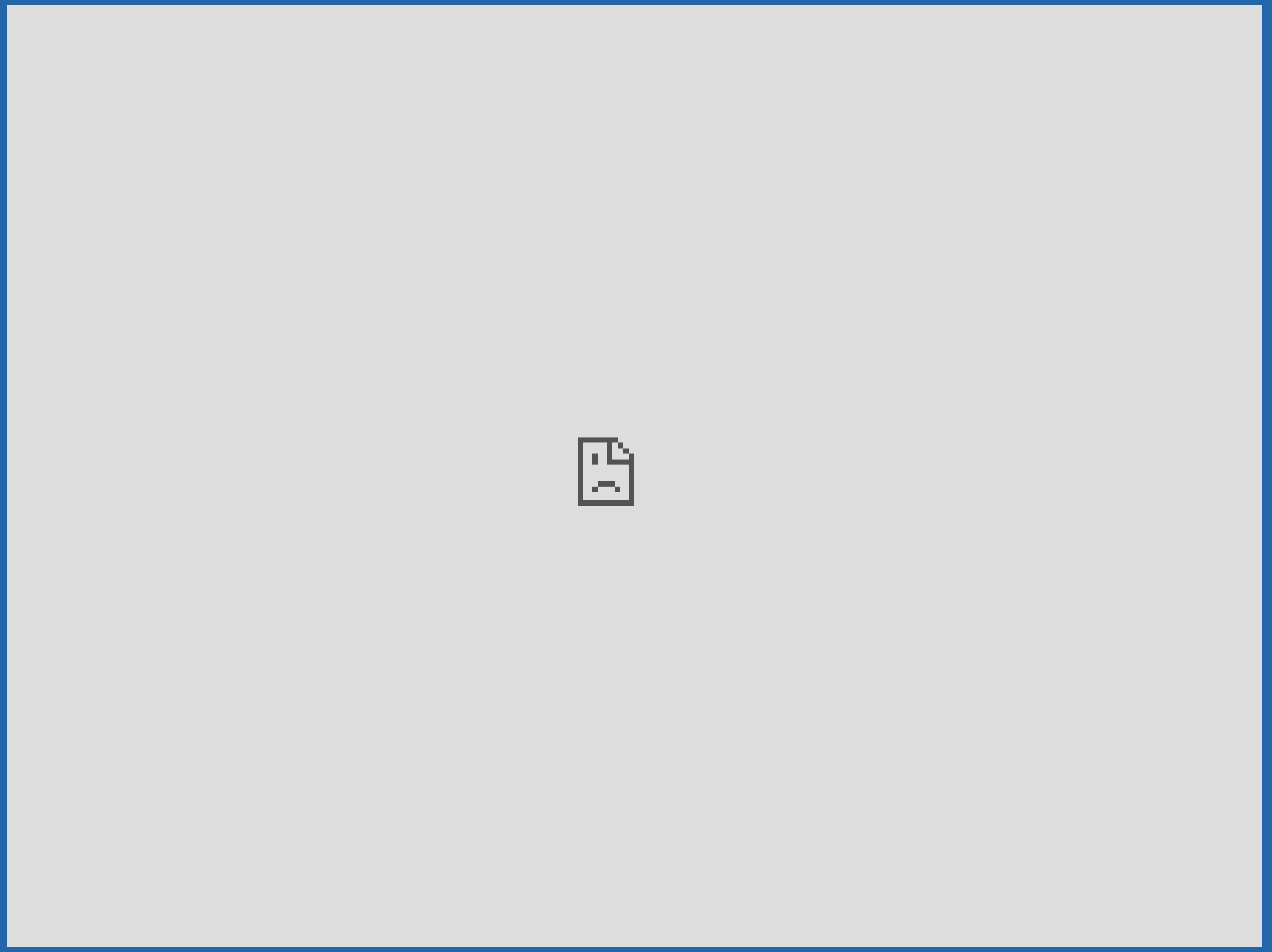


"I aware of it."



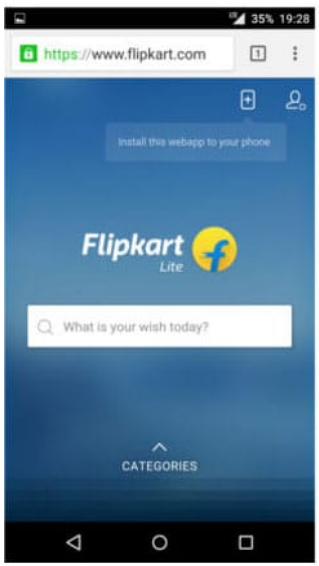


"This is a **bookmark**!"



[Youtube] Flipkart Lite Genius Ad.

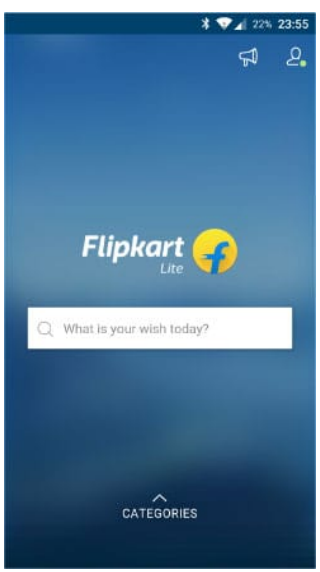
What user would think after teaching





"I can install App from Browser(URL)!"



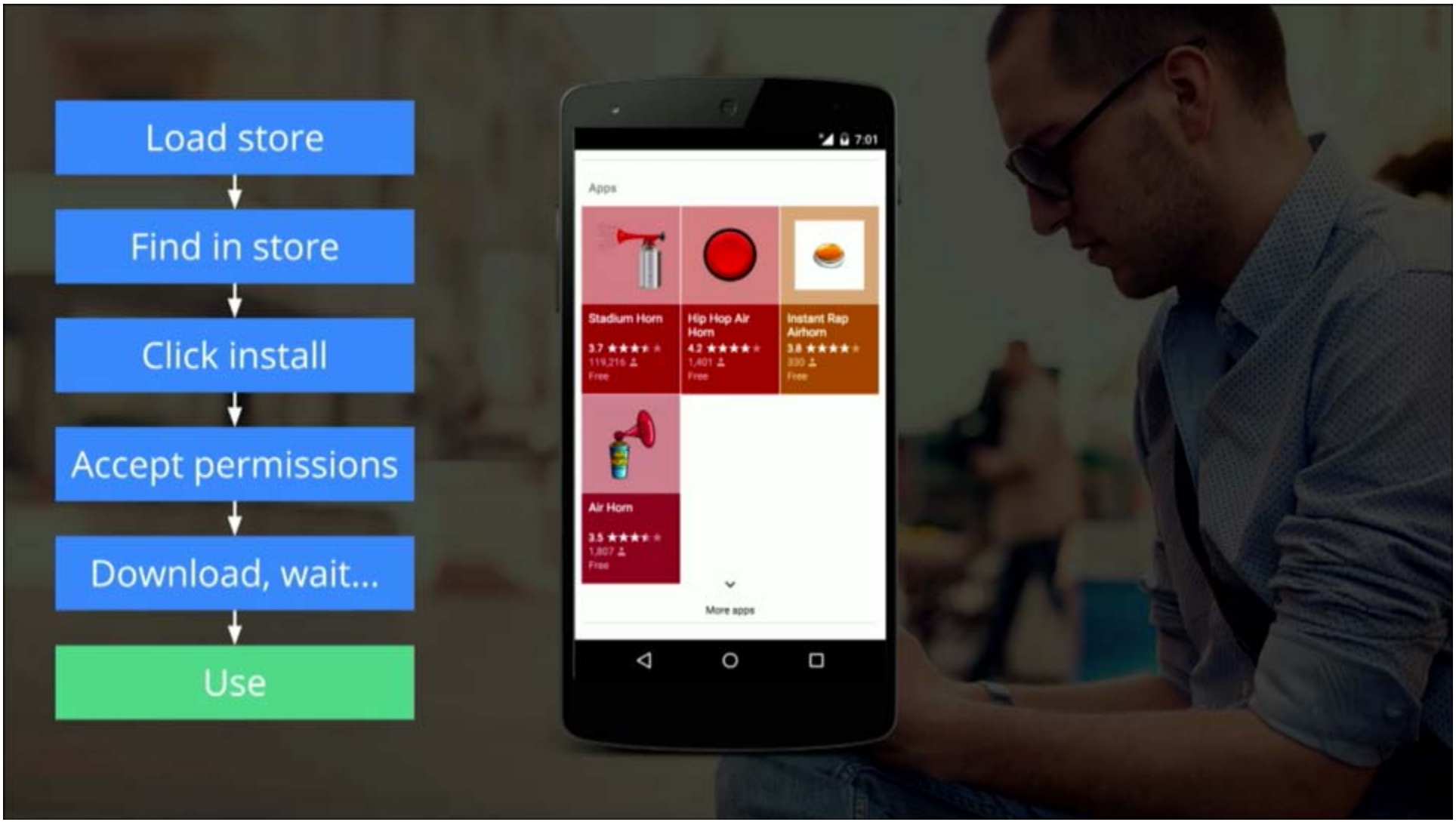




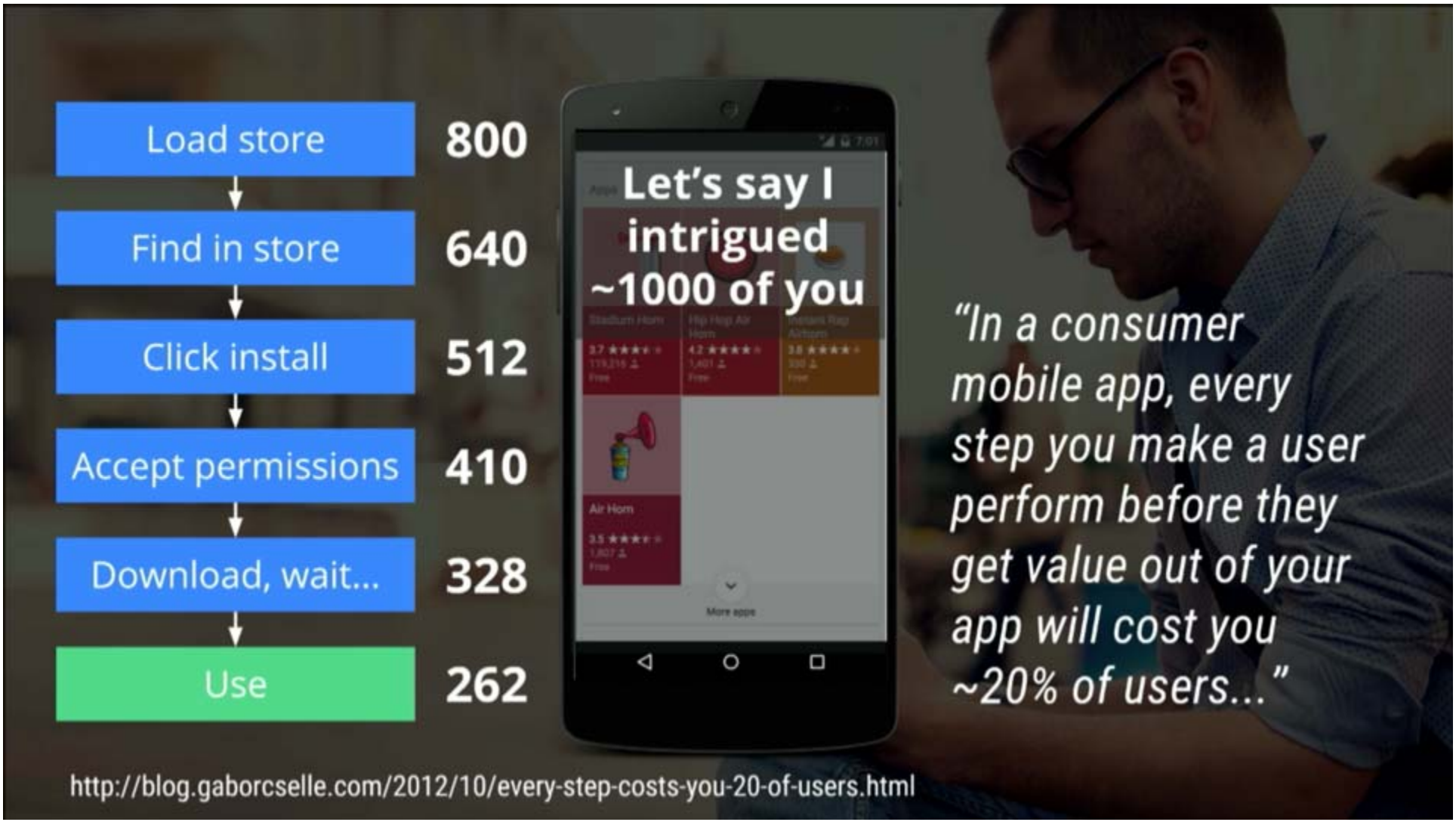
"This is an offline reliable **app**!"



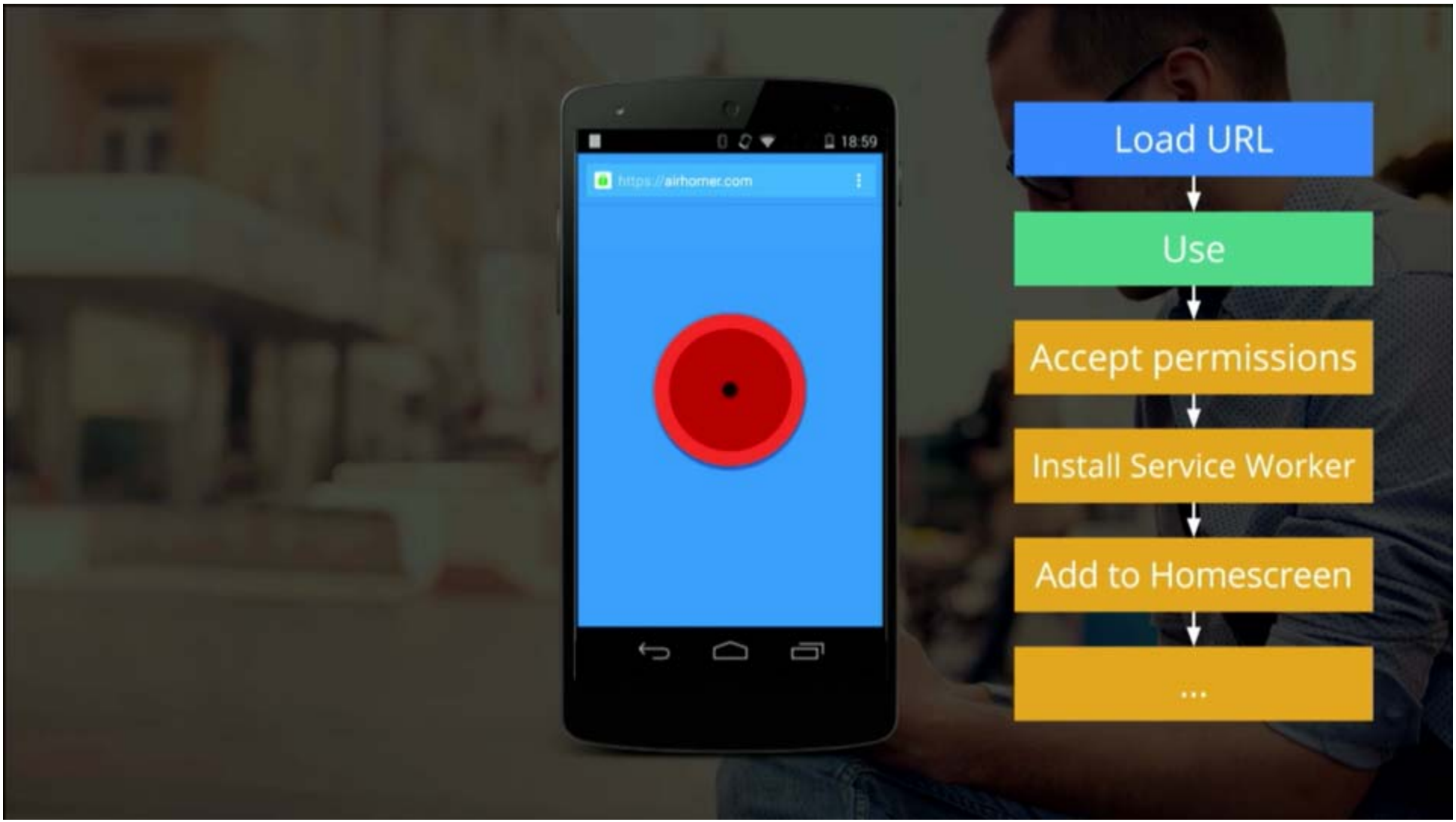
I FELT PRETTY CLEVER UNTIL I REALIZED I'D INVENTED WEBPAGES.



[Escaping tabs with progressive (web) apps - Ilya Grigorik - VelocityConf]



[Escaping tabs with progressive (web) apps - Ilya Grigorik - VelocityConf]



[Escaping tabs with progressive (web) apps - Ilya Grigorik - VelocityConf]

Progressive App

Streamable App

Evergreen App

Im Resistent To Answer This FAQ!

PWA vs. Others



“从纯 Web 到纯 Native，
之间有许多可能的点。”

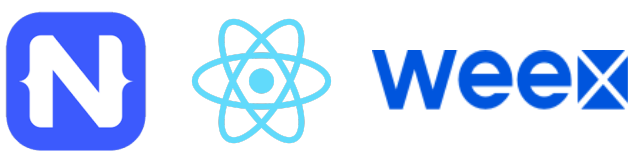
Low capability



Web



Hybrid



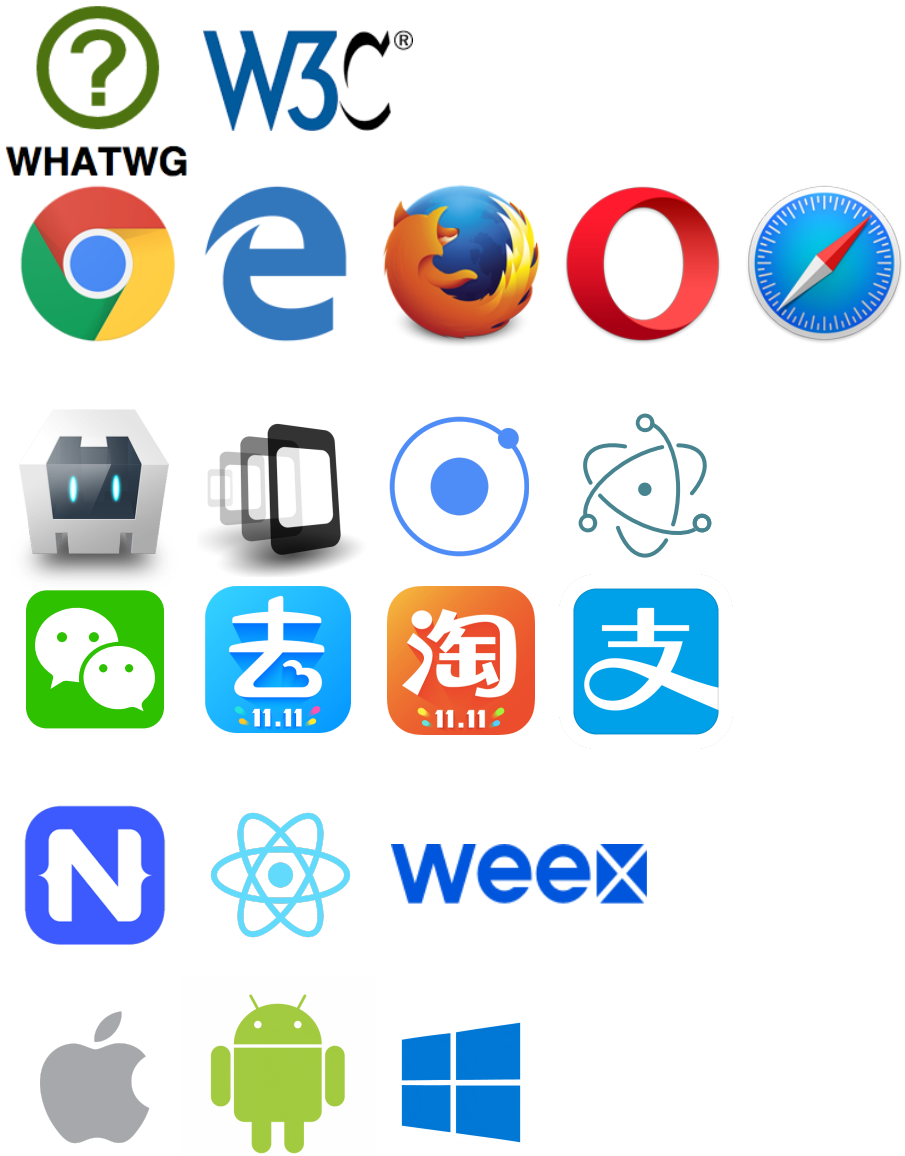
JS 2 Native



Native

High capability

Low capability



↑

Web

↓

↓

↓

Hybrid

↑

↑

JS 2 Native

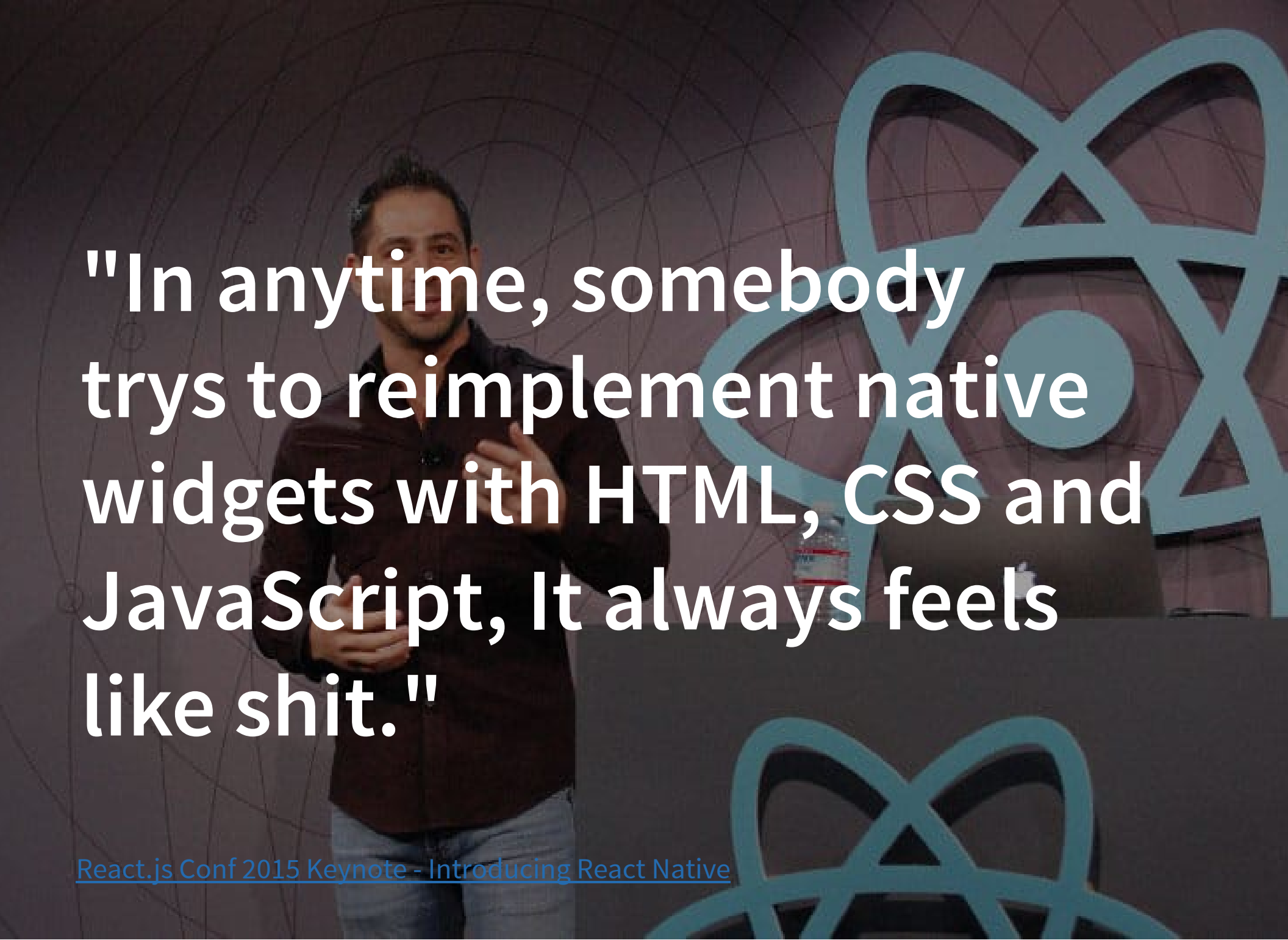
↓

Native

↓

High capability


娱票儿
票房
WE TRIED HYBRID AS WELL...
罗老师 别这样 别这样……罗老师……
No, Teacher Luo, come on...Teacher Luo...


"In anytime, somebody trys to reimplement native widgets with HTML, CSS and JavaScript, It always feels like shit."
React.js Conf 2015 Keynote - Introducing React Native

Web or Native

Compatibility Issue

Regional Issue

"Making the tradeoff for your product"

the-surprising-tradeoff-at-the-center-of-question-whether-to-build-an-native-or-web-app

"Every step on the path to a Progressive Web App makes sense on its own"

why-progressive-web-apps-vs-native-is-the-wrong-question-to-ask

https://cloudfour.com/thinks/progressive-web-apps-simply-make-sense/

It just make Web better.

It's Not Just About Software.

The Belief In Web



http://phonegap.com/blog/2012/05/09/phonegap-beliefs-goals-and-philosophy/


It's about "Anyone, at any time, can publish anything from anywhere. That is the stuff of revolutions, and our evolution as a species."
http://phonegap.com/blog/2012/05/09/phonegap-beliefs-goals-and-philosophy/



CC http://oreil.ly/tim_oneweb

We Choose A Hard Route, But We Are Damn Proud.

Nobody, maybe me, reply to the "Web as Adaptive Platform"


"We're alone in deciding that you should be able to run desktop applications(legacy) on a phone."
CODE
Google
Accessibility
whats-next-for-the-web-platform-alex-russell



[iPhone to Support Third-Party Web 2.0 Applications](#)
[WWDC 2007 - The Innovative new way to create apps for mobile](#)


Thanks
Blog / Weibo / GitHub / 知乎

Q/A

自豪的采用 @演说.io

huangxuan.me / Weibo / GitHub / 知乎



huangxuan.me/pwa-qcon2016

github.com/huxpro/pwa-qcon2016